Masked, Not Missing: Mid-Depth Retrieval Collapse in T5-Family Encoders

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Abstract

Practitioners often read intermediate layers of off-the-shelf encoders, check them for anisotropy with mean pairwise cosine, and correct the embedding space post hoc. We test this practice on medieval Latin canon law, where a scholarly edition labels which manuscript copies (witnesses) share a source. In the raw T5 encoders LaTa, PhilTa and mT5-base, 26 of 36 layers rank same-source pairs near chance, although their first and last layers retrieve well. Mean pairwise cosine gives no reliable warning of this collapse, while the share of variance on the first principal component is high at every collapsed layer. The signal is masked, not missing: a projection fit on training embeddings alone brings all 100 layers of our six-encoder panel to AUROC 0.962–0.987. Of the six raw T5-family encoders we test, four collapse: these three and T5-v1.1-base. All four share the T5 v1.1 layout; encoder-only siblings pretrained on the same corpora and two checkpoints with the original T5 layout do not. Removing about three principal directions restores every collapsed layer, while zeroing the ten largest coordinates restores none. After the projection the six encoders route witnesses to their source or to "new" about equally well, but no frozen encoder at its train-selected layer beats a character n-gram reference.

1 Introduction

Off-the-shelf encoders are a common tool for finding reused text in digital-humanities collections (e.g., Manjavacas et al., 2019; Schelb et al., 2026). Practitioners often read an intermediate layer, because intermediate layers can outperform the final one (Skean et al., 2025). They check the space for anisotropy, usually with the mean cosine between embeddings (Ethayarajh, 2019). They then correct it, for example by removing the top principal components (Mu and Viswanath, 2018) or by whitening (Su et al., 2021). This practice assumes that the check warns when a layer retrieves poorly. Mean cosine is already known to be a poor isotropy measure (Rudman et al., 2022); we ask whether it, or any label-free statistic, warns where retrieval fails.

Medieval Latin suits this test. The Carolingian Canon Law project (CCL; Firey, 2009) keys each transcribed manuscript unit to the source it copies, with expert labels that no model shaped. Latin also offers T5 and encoder-only models from the same authors and pretraining corpora. Our testbed holds 1,705 *witnesses*, manuscript copies of 840 sources. We evaluate two tasks: *ranking* pairs of witnesses by cosine, scored by AUROC, and *routing* each witness to its source or to "new". Because witnesses copy their source's wording, a character n-gram reference ranks pairs at 0.987 AUROC, so a retrieval failure points to the representation.

Three raw T5 encoders, trained with no embedding objective, fail at mid-depth (Figure 1). In LaTa, PhilTa and mT5-base (Xue et al., 2021), 26 of 36 layers rank same-source pairs at AUROC 0.50 to 0.66, near chance. Their first and last layers still retrieve. Three embedding-trained encoders, which complete our six-model panel, never collapse. The usual check, mean pairwise cosine, gives no reliable warning in either direction. The share of variance on the first principal component (top-PC share) is high at every collapsed layer: necessary for collapse in our pretrained models, but not sufficient.

The signal is masked, not missing. All-but-the-Top (ABTT; Mu and Viswanath, 2018), a linear projection fit on training embeddings only, removes the mean and the top principal components from every embedding, with no parameter update. It brings all 100 panel layers to AUROC 0.962 to 0.987, the 26 collapsed layers included. Because the projection sees no test label, the ranking it recovers is already present in each collapsed layer.

Among the six raw T5-family encoders we test, the collapse tracks the T5 v1.1 layout: a gated-

GELU feed-forward block (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020). LaBERTa and PhilBERTa, encoder-only siblings of LaTa and PhilTa, never fall below 0.826 and 0.883. T5-v1.1-base collapses on the same Latin text (lowest AUROC 0.489), while the original T5-base (Raffel et al., 2020) never falls below 0.816. T5-base also saw supervised tasks in pretraining, so we add T5-efficient-base (Tay et al., 2022), with T5-base's layout but pretrained on C4 alone, without dropout and on about 30 times fewer tokens; it does not collapse either. Of the six raw T5-family encoders we test, the four that collapse all have this layout. We do not isolate which change matters.

Inside the collapsed layers, about three directions mask the signal. Removing three principal components brings all 26 layers to AUROC 0.914 or above, and mT5-base keeps improving up to ten. Zeroing the ten largest coordinates restores none of them, although the first principal component (the dominant direction) sits mostly on such coordinates. In LaTa, pooling without the comma, the period and </s> restores every collapsed layer; in mT5-base, dropping up to its top 100 token types does not.

This paper makes four contributions. (1) A per-layer retrieval profile of T5 encoder states, the first we know of; it shows a mid-depth collapse that mean cosine misses, top-PC share flags and ABTT reverses (Section 4). (2) Controls showing that the collapse tracks the T5 v1.1 layout: T5-v1.1-base collapses where the original T5-base does not, encoder-only siblings on the same corpora do not, and the high-share middle layers recur on English text (Section 5). (3) A localization of the mask to about three directions, which zeroing the ten largest coordinates does not remove, with a token-level account in LaTa and none in mT5-base (Section 6). (4) A measure of what the projection gains for routing and where it stops, set against a character n-gram reference (Section 7). We will release the frozen testbed, split, evaluation code and character n-gram reference scores. We localize and repair the collapse but do not explain why it tracks the T5 v1.1 layout.

2 Related Work

Anisotropy statistics. Contextual embeddings occupy a narrow cone, usually checked by mean cosine over random pairs (Gao et al., 2019; Ethayarajh, 2019), which is not an appropriate measure of isotropy (Rudman et al., 2022). For pooled output embeddings, the variance share of the largest coordinate or top principal direction predicts how much rank-based similarity gains over cosine (Parupudi, 2026), and a topological signature led by effective rank moderately predicts retrieval (Rottach et al., 2025). Yet geometric statistics do not reliably predict downstream performance (Nastase and Merlo, 2025; Kulkarni et al., 2026). We relate mean cosine and top-PC share to retrieval at every layer (Section 4).

Outlier dimensions and their carrier tokens. One to three rogue dimensions dominate cosine similarity in pretrained transformers (Timkey and van Schijndel, 2021). In BERT they are tied to a few LayerNorm weights and to token frequency (Kovaleva et al., 2021; Puccetti et al., 2022). Clipping them improves mean-pooled sentence embeddings (Luo et al., 2021), though multilingual BERT is anisotropic without them (Rajaee and Pilehvar, 2022). Most BERT outliers sit on [SEP], the period and the comma (Bondarenko et al., 2023), and removing punctuation and special tokens before pooling improves sentence similarity (Fuster Baggetto and Fresno, 2022). Decoder language models carry massive activations on starting and delimiter tokens (Sun et al., 2024). The original T5's encoder carries outliers that grow with depth (Zhao et al., 2025), and outliers persist through fine-tuning (Rudman et al., 2023). On pooled T5 encoder states we compare coordinate zeroing, direction removal and token dropping (Section 6).

Layerwise analysis and post-hoc correction. Prior T5 embedding work reads the final layer (Ni et al., 2022) or the first and last layers averaged (Huang et al., 2021); Godey et al. (2024) trace anisotropy through T5-base's encoder and decoder layers but do not measure retrieval. Large decoder language models dip at mid-depth in retrieval and similarity and recover near the top, while two decoder embedding models stay roughly flat there before rising (Rizwan et al., 2026). Post-hoc corrections remove the mean and top principal components (Mu and Viswanath, 2018), whiten (Su et al., 2021), or down-weight frequent tokens (Arora et al., 2017). Timkey and van Schijndel (2021) apply them at every layer of four models and find that rogue dimensions obscure representational quality; our reading of the collapse as masked follows theirs, and we use ABTT as a probe. Whiten-

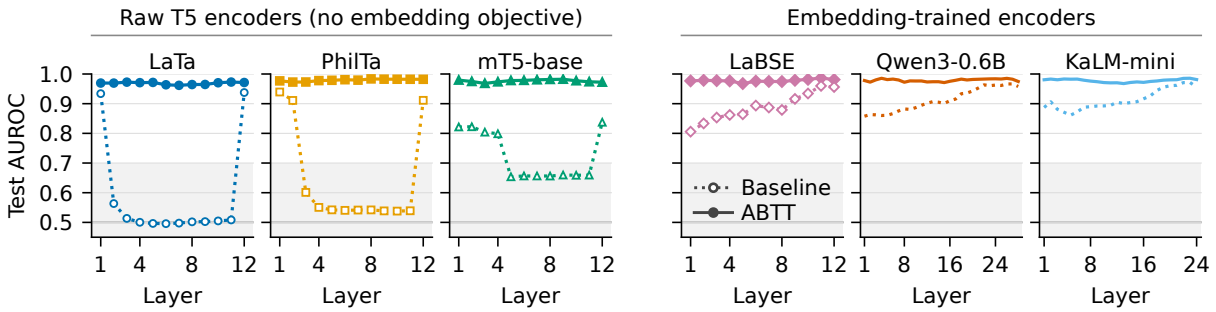


Figure 1: Test ranking AUROC at every layer of the six panel models, without correction (Baseline) and after ABTT, fit on training embeddings only. Gray band: collapsed (AUROC below 0.70); gray line: chance. The three raw T5-family panel models collapse at mid-depth; after ABTT every layer scores 0.962 to 0.987.

ing beats zeroing outlier dimensions in a multilingual encoder (Hämmerl et al., 2023). Centering alone gives embedding models gains under one point (Ren et al., 2026), and isotropy corrections stop helping after fine-tuning (Rajaee and Pilehvar, 2021). For routing, the nearest precedent is out-of-distribution detection in supervised classifiers, where low-variance components carry much of the signal (Kamoi and Kobayashi, 2020; Podolskiy et al., 2021), and nearest-neighbor scores depend on feature geometry (Sun et al., 2022). Lexical methods match or beat dense retrievers on Latin intertextuality (Schelb et al., 2026), while dense encoders reportedly help on paraphrastic reuse (de la Selle, 2026).

3 Experimental Setup

Corpus and labels. The testbed comes from CCL (Firey, 2009), an open-access edition of early medieval, mostly ninth-century, canon-law manuscripts. A *witness* is one manuscript's copy of a source text, kept as one file. Witnesses that copy the same source share a *directory*; the testbed holds 1,705 witnesses in 840 directories. The labels are expert-assigned by the project and checked by collation in its parallel display (Appendix A).

Split and train-only rule. We split the witnesses into 847 training and 858 test witnesses, with 565 and 596 same-source pairs; pairs form only within a split (Appendix B). The layer, routing threshold and ABTT are chosen or fit on the training split. The test split selects nothing.

Tasks and metrics. *Ranking* scores every pair of test witnesses by the cosine of their mean-pooled embeddings. We report AUROC with same-source pairs as positives. *Routing* is open-set: each test witness goes to an existing directory or to "new", and DirAcc@1 is the share of correct decisions (Section 7 gives the procedure).

Model	Type	L	$\text{AUROC}_{\min}$ (ℓ)	$\text{PC1}_{\max}$	Coll.
Raw T5 encoders, T5 v1.1 layout					
LaTa	T5	12	0.496 (6)	0.952	10
PhilTa	T5	12	0.538 (10)	0.858	9
mT5-base	T5	12	0.654 (5)	1.000	7
T5-v1.1-base	T5	12	0.489 (2)	0.975	10
Raw T5 encoders, original layout					
T5-base	T5	12	0.816 (11)	0.545	0
T5-efficient-base	T5	12	0.736 (11)	0.627	0
Raw encoder-only siblings					
LaBERTa	Enc.	12	0.826 (1)	0.254	0
PhilBERTa	Enc.	12	0.883 (6)	0.162	0
Embedding-trained					
LaBSE	Enc.	12	0.806 (1)	0.524	0
Qwen3-0.6B	Dec.	28	0.858 (1)	0.183	0
KaLM-mini	Dec.	24	0.861 (5)	0.419	0

Table 1: The eleven main-text models, read as mean-pooled hidden states with no correction. Type: T5 encoder (T5), encoder-only (Enc.) or decoder (Dec.). L: blocks read. $\text{AUROC}_{\min}$: lowest test ranking AUROC over layers, at layer ℓ. $\text{PC1}_{\max}$: peak top-PC share of training embeddings. Coll.: number of collapsed layers. Only the four raw T5 encoders with the T5 v1.1 layout collapse.

A *collapsed layer* has *baseline* (uncorrected) test AUROC below 0.70. The cutoff only describes layers; it selects no reported number. Two label-free statistics, computed on the training embeddings, describe each layer. The *mean pairwise cosine* is the usual anisotropy check (Ethayarajh, 2019). The *top-PC share* is the share of variance on the first principal component of the centered embeddings.

Models. Of the eleven models in Table 1, six form the *panel*, analyzed in full over 100 layers. Three are raw T5 encoders with no embedding objective: the Latin LaTa (Riemenschneider and Frank, 2023d), its Ancient Greek, Latin and English sibling PhilTa (Riemenschneider and Frank, 2023a), and mT5-base (Xue et al., 2021). Three are embedding-trained: LaBSE (Feng et al.,

2022), Qwen3-Embedding-0.6B (Qwen3-0.6B; Zhang et al., 2025) and KaLM-mini (Hu et al., 2025). The other five are raw: T5-efficient-base and four controls, each matched to another model in Section 5: the encoder-only siblings LaBERTa (Riemenschneider and Frank, 2023c) and PhilBERTa (Riemenschneider and Frank, 2023a), and the English T5 encoders T5-v1.1-base (Google Research, 2020) and the original T5-base (Raffel et al., 2020). We mean-pool the hidden states of every transformer block, dropping only padding and tokens that decode to no text, such as SentencePiece's bare word-boundary piece; special tokens such as </s> and punctuation are kept. Appendix H compares last-token pooling. Table 5 lists all thirteen models, adding Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b).

Post-processing. ABTT (Mu and Viswanath, 2018) subtracts the training mean and removes the top D principal components of the centered training embeddings from every vector. We choose $D \in \{1, 2, 3, 5, 7, 10\}$ per layer by training DirAcc@1. ABTT is fit on training embeddings only and updates no parameter; we also call it *the projection*. We compare SIF frequency weighting (Arora et al., 2017) and reduced-rank whitening (Su et al., 2021) in Appendix I.

4 Collapse by Depth

In LaTa, PhilTa and mT5-base most middle layers rank near chance. Mean cosine gives no reliable warning, but top-PC share is high at every collapsed layer. ABTT restores ranking at every panel layer.

LaTa, PhilTa and mT5-base fall to near chance at mid-depth. Figure 1 gives baseline ranking AUROC at every layer of the six panel models. In LaTa, PhilTa and mT5-base, 26 of the 36 layers collapse, ranking at AUROC 0.50 to 0.66. They are layers 2–11 of LaTa, 3–11 of PhilTa and 5–11 of mT5-base. The lowest remaining panel layer ranks at 0.799 (mT5-base, layer 4), so any cutoff from 0.66 to 0.79 gives the same 26 layers. Training AUROC below 0.70 picks the same 26 layers. The first and last layers still retrieve; LaTa ranks at 0.934 at layer 1 and 0.938 at layer 12. The embedding-trained panel models never rank below 0.806.

A train-only projection restores ranking at every layer of the six panel models. ABTT (Section 3)

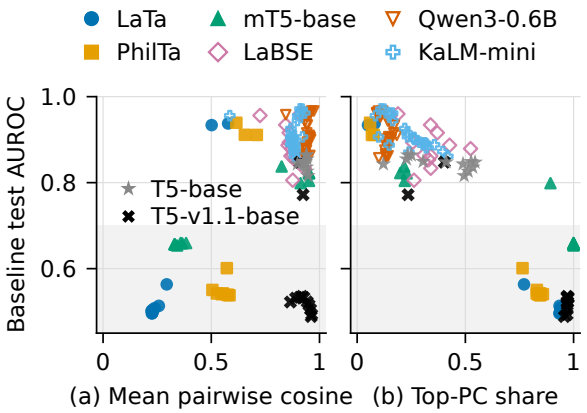


Figure 2: Baseline test AUROC against (a) mean pairwise cosine and (b) top-PC share of the training embeddings, one point per layer of the six panel models, T5-base and T5-v1.1-base. Filled: raw T5-family; open: embedding-trained. Gray band: collapsed (AUROC below 0.70). Collapsed layers span low and high mean cosine; all have a top-PC share of at least 0.76, as does one healthy layer.

lifts all 100 layers of the panel, the 26 collapsed layers included, to AUROC 0.962 to 0.987 (Figure 1, solid lines). Reduced-rank whitening (128 or 256 components), fit on the same training embeddings, ranks within 0.021 of ABTT at every panel layer (Appendix I). The narrow band holds for ranking only. Routing after ABTT still varies by layer, from 75.6 to 90.2 DirAcc@1 (Appendix H).

Mean pairwise cosine gives no reliable warning. Figure 2a plots training-set mean pairwise cosine against baseline test AUROC. Collapsed panel layers have a lower mean cosine (0.22 to 0.58) than any embedding-trained panel layer (0.59 to 0.97). But LaTa's healthy layers 1 and 12 sit in the collapsed range (0.50 and 0.58). T5-v1.1-base, which also collapses (Table 1; Section 5), sits at 0.87 to 0.96 at every layer, collapsed or not. Neither a high nor a low mean cosine marks a collapsed layer.

Top-PC share is high at every collapsed layer of the pretrained models. Top-PC share needs no labels either (Figure 2b). It is at least 0.76 at every collapsed layer of the panel and of T5-v1.1-base. In the three embedding-trained panel models it peaks at 0.524 (LaBSE). A high share is necessary for collapse in our pretrained models but not sufficient: of the 88 healthy layers in Figure 2, only mT5-base layer 4 reaches 0.76 (0.893, AUROC 0.799). Sentence-T5 (Ni et al., 2022), an embedding-trained T5 encoder, has a share of 0.76 or more at layers 2–9 (peak 0.94) yet ranks at 0.800 or above at every layer (Table 5). Among collapsed

layers a higher share does not predict a lower AUROC (Appendix C).

5 Collapse Across Models

Of the eleven models in Table 1, four collapse: LaTa, PhilTa, mT5-base and T5-v1.1-base. All four have the *T5 v1.1 layout*: a gated-GELU feed-forward (Shazeer, 2020) and untied input and output embeddings (Google Research, 2020) (Table 5). None of the other seven collapses at any layer. T5-base and T5-efficient-base keep a ReLU feed-forward and tied embeddings.

Encoder-only siblings pretrained on the same corpora do not collapse. LaBERTa and PhilBERTa share authors and pretraining corpora with LaTa and PhilTa (Riemenschneider and Frank, 2023d,c,a) and, like them, have no embedding objective. LaTa and PhilTa fall to 0.496 and 0.538 AUROC; the siblings never fall below 0.826 (LaBERTa) and 0.883 (PhilBERTa). Their top-PC share peaks at 0.254 and 0.162 (vs. 0.952 and 0.858 in LaTa and PhilTa). Shared pretraining text and a missing embedding objective do not produce the collapse on their own. Architecture, layout included, pretraining objective and tokenizer vary together within each pair, so the pairs cannot say which matters (Appendix D).

T5-v1.1-base collapses; T5-base and T5-efficient-base, with the original layout, do not. T5-v1.1-base and T5-base share their tokenizer, the span-corruption objective and C4 pretraining text (Raffel et al., 2020; Google Research, 2020). On Latin text, T5-v1.1-base falls to 0.489 at layer 2, and its layers 2 to 11 collapse. T5-base stays at 0.816 or above at every layer, with top-PC share peaking at 0.545. The two differ in layout and in pretraining mix: T5-base also saw supervised tasks (Raffel et al., 2020). To separate layout from that mix we add T5-efficient-base (Tay et al., 2022): T5-base's layout, but pretrained like T5-v1.1-base on C4 alone and without dropout, and on about 30 times fewer tokens than either. At that budget it does not collapse: its lowest AUROC (0.736) and its peak top-PC share (0.627) both fall at layer 11 (Table 5). On the four controls in Table 1, ABTT lifts every layer to AUROC 0.927 or above, the ten collapsed layers of T5-v1.1-base included, but eight of the 48 (seven in T5-v1.1-base) stay below the panel's band (0.962–0.987; Table 6). Among the six raw T5-family encoders we test, the collapse tracks the T5 v1.1 layout; we do not isolate which change is responsible.

In three models with the T5 v1.1 layout, high top-PC share in the middle layers appears on English text too. T5-v1.1-base was pretrained on English alone, so its high-share layers on Latin could reflect a language mismatch. We read mT5-base, PhilTa and T5-v1.1-base on 847 English passages of US court opinions (Harvard Law School Library Innovation Lab, 2024; Kandpal et al., 2025), paired with the Latin training witnesses and matched in mT5 token length. On English, in all three models, layers 4 to 11 have a top-PC share of at least 0.82 and the first and last layers stay at or below 0.34 (Figure 5). At layers 2 to 11 of T5-v1.1-base, top-PC share is 0.931 to 0.950 on English and 0.958 to 0.975 on Latin. The check covers geometry only; without labels, it says nothing about English retrieval.

In LaTa, contrastive fine-tuning leaves the middle layers collapsed. In a small test, we fine-tune LaTa on 499 training pairs, adding the contrastive embedding objective it lacks (Appendix G). Fine-tuning lifts the last layer from 0.938 to 0.984 AUROC but leaves layers 2 to 11 at 0.50 to 0.57. Peak top-PC share barely moves (0.952 before, 0.945 after; Table 7). Hämmerl et al. (2023) report the same split: their sentence-tuned multilingual encoder keeps a large outlier dimension at a middle layer but none at its output layer.

6 Localizing the Collapse

In the collapsed layers about three directions mask the retrieval signal. Zeroing the ten largest coordinates does not remove this mask, and a few token types feed it in LaTa but not in mT5-base.

We run each test at the 26 collapsed layers, fit every intervention on training embeddings, and read test ranking AUROC. A layer counts as restored at AUROC 0.90 or above. Appendix E gives further tables and controls.

Zeroing the ten largest coordinates restores no collapsed layer. In pretrained transformers a few outlier coordinates dominate cosine similarity (Timkey and van Schijndel, 2021). Zeroing them improves pooled similarity in pretrained encoders (Luo et al., 2021), and whitening does better still (Hämmerl et al., 2023). In our encoders, zeroing up to ten coordinates, ranked on training embeddings by variance or by mean absolute value, restores

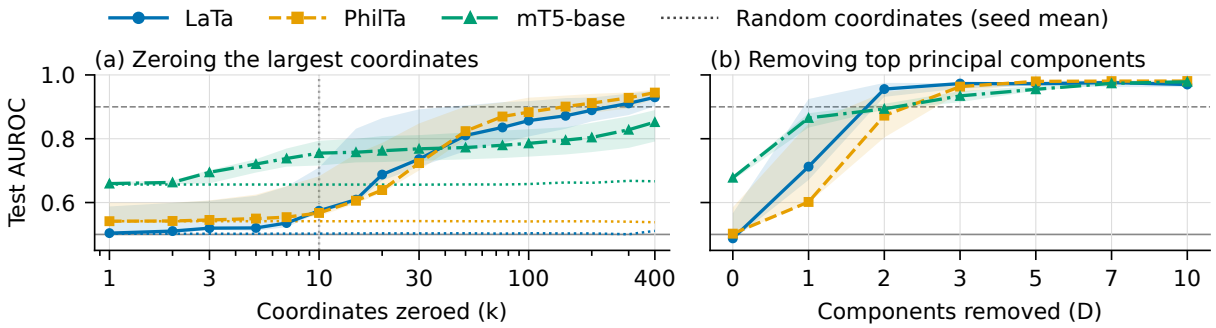


Figure 3: Test ranking AUROC at the 26 collapsed layers of LaTa (10 layers), PhilTa (9) and mT5-base (7). Lines and bands: median and range per model; dashed line: AUROC 0.90; gray line: chance. (a) Zeroing the k coordinates of largest mean absolute value: no layer passes 0.90 by $k = 10$ (vertical line). (b) Removing the top D principal components (ticks evenly spaced, D values not): from $D = 3$ every layer passes 0.90.

none of the 26, and the best layer reaches 0.794 (Figure 3a). The dominant direction loads mostly on these coordinates: the ten of highest variance carry a median 92% of its squared loading. Yet after zeroing them, its remainder still holds a median 72% of the remaining variance. Zeroing far more coordinates does restore LaTa and PhilTa. Ranked by variance, recovery starts at 50 coordinates, and their 19 collapsed layers all recover by 300 (by 400 under the mean-absolute-value ranking of Figure 3a). No mT5-base layer recovers even at 400. The dominant direction loads mostly on a few coordinates but acts beyond them, which extends the outlier-dimension account.

Removing three directions restores every collapsed layer. ABTT removes whole directions instead of coordinates, so we vary the number D of principal components it removes (Figure 3b). Centering alone ($D = 0$) does not help: the median AUROC over the collapsed layers falls from 0.541 to 0.503. At these layers the first component holds a median 94% of the centered training variance. It is not mainly a passage-length direction: $|\rho| \leq 0.44$ with log token count at all 26 layers. Removing it alone recovers a median 45% of the AUROC gain from removing ten components, and 80% or more at only one of the 26 layers. Removing three components brings all 26 layers to 0.914 or above. mT5-base keeps improving up to ten components (median 0.934 at three, 0.978 at ten). The ten removed components rank at 0.489, the next ten at 0.919, and the retained subspace at 0.977 (Table 10). At these layers the signal lies outside the removed directions.

In LaTa, three token types feed the removed directions. A pooled vector is the mean of its token vectors, so a few token types with large values along the removed directions could dominate a witness's scores on those directions. We rank token types by their share of the variance of training-witness scores on the top three components, then pool without the top m types, for m up to 100. We ask whether some m restores the collapsed layers. In LaTa, $m = 3$ does. Pooling without the comma, the period and `</s>` drops 11% of the test tokens and restores all 10 collapsed layers (median 0.919). Random token types of the same training share leave the median at 0.500. The tokens are carriers, not a cause, since LaBERTa also pools commas, periods and `</s>` and does not collapse (Section 5). PhilTa needs more types: dropping its top 30, about a third of its tokens, restores all 9 of its collapsed layers on test passages (settings chosen per layer on training AUROC restore every collapsed layer of both models, medians 0.926 and 0.928).

In mT5-base, dropping up to its top 100 token types restores no collapsed layer. Dropping up to 100 token types, ranked the same way, restores none of its 7 collapsed layers, the best reaching 0.748. mT5-base also resists coordinate zeroing, yet removing three to ten directions repairs it. Its mask sits neither in a few coordinates nor in a few token types, and we have no token-level account.

7 Routing and Limits

At their train-selected layers after ABTT, the six frozen encoders route about equally well, but none beats a character n-gram reference.

Of the 858 test witnesses, 535 have a same-source partner in test (*existing*) and 323 do not (*new*). Each test directory scores a witness by the highest cosine between the witness and any other member. A pseudo-directory `NEW` scores a fixed threshold τ, chosen to maximize training F1 at

	Ranking AUROC		Routing DirAcc@1	
Model	Base	ABTT	Base	ABTT
LaTa	0.938 [.910,.963]	0.971$^{\downarrow}$ [.948,.991]	72.1 [68.4,75.8]	86.1$^{\downarrow}$ [83.2,88.9]
PhilTa	0.939 [.918,.958]	0.983 [.969,.993]	69.3 [65.2,73.5]	88.3 [85.6,90.9]
mT5-base	0.838 [.810,.866]	0.975 [.954,.992]	46.6 [42.5,51.0]	88.5 [85.7,91.2]
LaBSE	0.956 [.932,.977]	0.987 [.977,.995]	81.5 [78.1,84.7]	88.5 [85.8,90.9]
Qwen3-0.6B	0.966 [.950,.981]	0.973$^{\downarrow}$ [.948,.990]	80.3 [76.9,83.6]	89.4 [86.7,92.0]
KaLM-mini	0.972 [.955,.986]	0.981 [.963,.993]	85.9 [82.9,88.7]	89.4 [86.7,91.9]
Spread	0.134	0.016	39.3	3.3
Fine-tuned; Base = fine-tuned, ABTT = fine-tuned + ABTT				
LaTa	0.984	0.970	81.6	85.2
Qwen3-0.6B	0.996	0.994	90.8	91.4
KaLM-mini	0.997	0.994	91.7	92.5
Char. n-grams	0.987 [.976,.996]		89.9 [87.3,92.3]	

Table 2: Test ranking (AUROC) and routing (DirAcc@1, %), uncorrected (Base) and after ABTT, each at its train-selected layer (Table 16). Fine-tuned: contrastively fine-tuned encoders, intervals in Table 12. Char. n-grams: character 3–5-gram TF-IDF. Brackets: 95% bootstrap intervals over the 514 test directories. Spread: range over the six encoders. ↓: below the n-gram reference, paired interval excluding zero (Table 13); overlapping brackets do not imply a tie. ABTT cuts the routing spread from 39.3 to 3.3 points; no frozen encoder beats the reference.

separating same-source from different-source pairs. The witness goes to the highest-scoring directory. DirAcc@1 counts a decision as correct if it names the labeled directory, or NEW for a new witness. Assignment accuracy scores only the existing-versus-new decision; always answering existing scores 62.4%. Each headline number reads a model at its *train-selected* layer: the one of highest training AUROC for ranking, or of highest training DirAcc@1 for routing (Table 16). The task is lexically solvable by design, since the witnesses of one source copy its wording. Our *character n-gram reference*, TF-IDF over character 3–5-grams fit on training witnesses, reaches 0.987 AUROC and 89.9 DirAcc@1 (Table 12).

At train-selected layers, routing after the projection converges across the six encoders. At the train-selected layers of Table 2, the DirAcc@1 spread across the six encoders falls from 39.3 points without correction (95% CI 34.6 to 43.7) to 3.3 after ABTT (1.8 to 6.0). Uncorrected, they run from 46.6 (mT5-base) to 85.9 (KaLM-mini); after ABTT, from 86.1 to 89.4. Every encoder's routing gain has a bootstrap interval excluding zero, from 3.5 points for KaLM-mini (1.0 to 6.0) to 41.8 for mT5-base (Table 13).

For the embedding-trained encoders, the routing gain does not depend on the threshold. LaBSE, Qwen3-0.6B and KaLM-mini already rank well without correction (AUROC 0.956 to 0.972), so their routing gain cannot come from a ranking repair. ABTT still raises their DirAcc@1 by 7.0, 9.1 and 3.5 points. Because the gain could come from a better-placed threshold τ, we also score the existing-versus-new decision without one, as the AUROC of each test witness's best-match cosine, existing against new. ABTT raises it by 0.037 for LaBSE, 0.055 for Qwen3-0.6B and 0.027 for KaLM-mini, and every paired interval excludes zero (Appendix F; per-cell values in Table 14). ABTT's gain in assignment accuracy also holds at each setting's best test threshold and under a finer threshold grid (Appendix F). For these three encoders, centering alone ($D = 0$) scores 12 to 26 points lower in assignment accuracy than ABTT at the same layer (Table 13). We do not test what in the removed components produces this gain.

No frozen encoder at its train-selected layer beats a character n-gram reference. The best ABTT results in Table 2 tie the reference: LaBSE ranks at 0.987, and Qwen3-0.6B and KaLM-mini route at 89.4, with every difference interval including zero. Two ranking cells fall below it with intervals excluding zero, LaTa and Qwen3-0.6B (by 0.016 and 0.014 AUROC); mT5-base falls 0.012 below, its interval touching zero (0.000 to 0.027). LaTa's routing cell falls 3.7 points below it (Table 13). Our claims thus concern where the signal sits in the encoders and what masks it.

Fine-tuning Qwen3-0.6B and KaLM-mini adds about two routing points over the projection. As a supervised reference, we fine-tune LaTa, Qwen3-0.6B and KaLM-mini contrastively on 499 training pairs (Appendix G). For routing, fine-tuning plus ABTT beats zero-shot ABTT by 1.8 points for Qwen3-0.6B (92.3 against 90.5) and by 2.1 for KaLM-mini (93.0 against 90.9); on LaTa the two tie at 87.7. These are means over five re-seedings, each a new random draw of which test witnesses are queries and which are the references they are routed against (Table 15). In the single-split cells of Table 2 the gains are 2.0 and 3.1 points

(only KaLM-mini's interval excludes zero), and LaTa falls 0.9. These references are flattered: 206 of the 535 existing test witnesses (38.5%) sit in a directory that supplied a training pair. Fine-tuned Qwen3-0.6B and KaLM-mini rank above the n-gram reference in point estimate (0.996 and 0.997 AUROC). After fine-tuning, ABTT no longer helps ranking: AUROC falls slightly in point estimate in all three models (by 0.002 to 0.014; every interval includes zero), in line with Rajaee and Pilehvar (2021).

8 Discussion

Before reading an intermediate layer of a frozen encoder, fit a projection on training embeddings. In the six panel models it scored above the baseline in point estimate at all 100 layers in ranking and routing (Appendix H), which makes it cheap insurance. Routing after it still varies by layer, so the layer still matters. To screen uncorrected states, check top-PC share, not mean pairwise cosine. In our models mean cosine gave no reliable warning in either direction, while every collapsed layer of the pretrained models had a top-PC share of at least 0.76 (Section 4). The 0.76 is the lowest share we observed at a collapsed layer of a pretrained checkpoint, not a cutoff validated on held-out models. A high share is necessary for collapse here but not sufficient, and does not order layers inside the collapsed regime. Choosing among layers still needs a train-only task signal, such as training AUROC.

Why models with the T5 v1.1 layout collapse remains open. Gated feed-forward blocks are reported to amplify outlier activations in decoder language models (Fishman et al., 2025; Yang et al., 2025), but the original T5 also carries large outliers (Zhao et al., 2025). Our localization names carriers in LaTa but no cause. Concurrently, Mikkelsen (2026) reports a mid-depth retrieval trough on clinical text; retrieval-trained encoders recover in the last layers, and encoders without a retrieval objective recover only partly. In our models, by contrast, neither the embedding-trained encoders nor the encoder-only siblings collapse, and it tests no T5 model.

For the editor of a text collection, ABTT makes the choice among these six encoders matter little (Section 7). The existing-versus-new decision is still wrong for about one test witness in ten (assignment accuracy 88.5 to 91.7; Table 12). The final call stays with the scholar. Because the task is lexically solvable, a practical tool can start from character n-grams; dense encoders are reported to help where reuse is paraphrastic (de la Selle, 2026), a case we do not test.

Limitations

Four factors are uncontrolled. (i) We use one labeled corpus in one language and genre, early medieval Latin canon law; its 1,705 witnesses are many for this field but few by current NLP standards. (ii) Within each sibling pair, architecture, pretraining objective and tokenizer vary together, so the pairs do not say which of them matters. That LaTa and LaBERTa share authors and pretraining corpora rests on their model cards (Riemenschneider and Frank, 2023d,c). (iii) The T5 v1.1 changes (gated-GELU feed-forward with a narrower inner width, untied embeddings) vary together, so we do not say which of them matters. T5-efficient-base has the original layout without T5-base's supervised pretraining, but it is a single checkpoint pretrained on about 30 times fewer tokens; it separates layout from pretraining mix only at that budget. (iv) The token ablation accounts for the masking directions in LaTa with three token types and in PhilTa only with 30, about a third of its tokens; in mT5-base dropping up to its top 100 token types restores no collapsed layer, so its mask has no token-level account.

The scope of three results is narrow. We claim the narrow band for the six-model panel only: on the four controls and T5-efficient-base ABTT lifts every layer to AUROC 0.927 or above, but nine of their 60 layers, seven of them in T5-v1.1-base, stay below the panel's band (0.962 to 0.987). The English check measures geometry only; we have no labels for English retrieval. The task is lexically solvable by design, and a character n-gram reference ties the best encoders after ABTT, so we do not claim that encoders beat surface overlap on this corpus.

The fine-tuning test is small (499 pairs, at most eight epochs), and its references are flattered by directory overlap with the training pairs (Section 7). Our intervals resample test directories with the layer, threshold and ABTT fit held fixed; they cover test sampling and not the choice of split. ABTT needs enough text to estimate principal components, so smaller archives need separate study.

The labels are expert-assigned, but by one project. Its director or supervised graduate students assigned the keys, and verification is by collation in the project's parallel display, not by independent double annotation. No inter-annotator agreement figure therefore exists for this testbed. Collation needs at least two witnesses of a key, and 545 of the 840 directories hold only one in this testbed.

Ethical Considerations

The testbed holds early medieval legal texts, transcribed from manuscripts and early printings, and contains no personal data. The Carolingian Canon Law project publishes its material under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 license (CCL, 2026a). The project has confirmed in writing that the license covers the transcriptions, annotations and markup in the exports we use and that, in the view of the project's Advisory Board, attribution to the project suffices; the project's interface credits the transcriber and proofreader of each unit. The project's corpus keeps growing and being corrected, so the testbed is a dated snapshot (version 1), fixed by a sha256 manifest of its 1,705 labeled files, and every number in this paper comes from it. We will release it under the same license, in coordination with the project's periodic versioned exports, and later versions will follow those exports. The intended use is research on retrieval and on assisted source identification: the system ranks candidates for an expert to judge and does not identify sources on its own. The English control sample consists of public-domain US court opinions from the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), which name litigants; we use it only for label-free geometry statistics and do not release it. We used AI assistants for writing and checking code; the authors verified all code and numbers.

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A Corpus Provenance and Derivation

This appendix records where the testbed text comes from and how our files were derived from it, in the detail that Section 3 leaves out.

The project. CCL (Firey, 2009) is an open-access digital edition founded in 2009; Eichbauer

(2014) reviews it.[1] Its texts are early medieval canon-law manuscripts, most of them ninth-century and never edited to modern standards. A witness is one manuscript's copy of a source text, transcribed as one unit (one canon or excerpt). Transcription is manual, from manuscript images or, for some texts, eighteenth-century printings, except for one manuscript transcribed by handwritten text recognition (Paris, BnF, lat. 1454; 142 of the 1,705 witnesses), which is still in proofreading. Since 2019, a proofreader checks each new transcription against an image of the folio, letter by letter (CCL, 2026b).

Labels. The project has two identifier systems, both designed at its launch. Our file names are location identifiers: they record the manuscript siglum, the folium and the unit's position on it, and encode no source. The source key names the common source that units in different manuscripts copy, such as a fourth-century council or a papal decretal, once that source is identified, and our directories reuse that key as supplied. A key encodes the type of source, its point of origin, its date and its reference in modern scholarship, such as a canon number. A label counts as key-named when it begins with a four-letter code, a period, a year and a period; by this rule 689 of the 840 directory labels are source keys. The other 151 are modern edition citations, the Apostolic Canons, references to other collections, biblical references, and a handful of unidentified sources; each unidentified source is a singleton directory and contributes no positive pairs.

Assignment and verification. Keys were assigned by the project's director, a specialist in early medieval canon law, or by graduate students working under the director's supervision, who had enough Latin to check a manuscript version against the modern authoritative version of the source. Verification, and occasional correction, runs through the project's parallel display, which sets side by side all units that share a key and shows how far their wording agrees. A misidentified unit stands out there as a text that cannot be a version of the others. This is verification by collation (see Limitations). The project supplied two label corrections during this study, and both are logged with the frozen testbed. We assigned no label ourselves and changed only those two. What remains unidentified is large: across the several hundred manuscripts that contain the project's texts, thousands of units still carry no key, and they are the queries that routing is meant to serve.

Derivation. We hold the witnesses as plain text derived from CCL TEI-P5 exports. The rule is one file per unit: the directory is the source key from the unit's `scholSource` division, which is stripped from the text so that no file carries its own label; the text is the manuscript rubric followed by the canon text, with inline markup flattened (editorial notes kept, deletions dropped) and layout whitespace collapsed; a unit without a source key receives no directory. The rule is scripted. On a current export of one manuscript (Paris, BnF, lat. 2123), it reproduces all 279 units present in both the export and the corpus character for character after whitespace normalization, and places every one in the same directory.

B Experimental Setup Details

This appendix gives the setup details that Section 3 leaves out.

Split. Singleton directories (545 of 840) are divided file by file between the two splits. Doubletons go whole to one side. Larger directories are divided internally, so that every size class yields training positives. Pairs form only within a split.

KaLM-mini attention. KaLM-mini, a decoder, attends bidirectionally, as its checkpoint's own modeling code specifies.

Further layer statistics. Let $p_i = \lambda_i / \sum_j \lambda_j$ for the covariance eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots$ of the centered training embeddings; the top-PC share is p_1. The *effective rank* is $\exp(-\sum_i p_i \log p_i)$, the entropy effective rank of Roy and Vetterli (2007) applied to the covariance eigenvalues. It is 1 when one direction holds all the variance. The *cosine gap* g is the mean same-directory test cosine minus the mean different-directory test cosine. The *standardized gap* g/s divides it by the standard deviation s of all pairwise test cosines, which puts models with different cosine spreads on one scale. The two gaps need labels; the effective rank, like the top-PC share and the mean pairwise cosine, does not. Our mean pairwise cosine averages over all pairs of pooled passages, an adaptation of the word-level

[1] Founding year, proofreading standard, the one manuscript transcribed by handwritten text recognition, the design of the identifiers, the assignment and verification of keys, and the extent of unidentified material: unpublished documentation and correspondence supplied by the project, 2026.

statistic of Ethayarajh (2019), which uses random pairs.

SIF. SIF (Arora et al., 2017) reweights each token by $a/(a + p(w))$ before pooling, with $a = 10^{-3}$ and $p(w)$ estimated on training texts. It gives special tokens, such as the end-of-sequence token `</s>`, zero weight. We use only this weighting step of SIF, not its common-component removal. SIF+ABTT applies both.

Software. We extract hidden states with PyTorch (Paszke et al., 2019) and Hugging Face Transformers (Wolf et al., 2020), fit whitening and the character TF-IDF reference with scikit-learn (Pedregosa et al., 2011), and compute principal components and the remaining statistics with NumPy (Harris et al., 2020). BM25 is our own implementation.

C Collapse by Depth: Further Diagnostics

This appendix adds the diagnostics behind the four findings of Section 4 and the SIF baseline, on baseline mean-pooled hidden states unless a paragraph says otherwise. Per-layer values are in the code release.

Geometry at the collapsed layers. At the 26 collapsed layers, effective rank lies between 1.0 and 4.6; in the embedding-trained models it ranges from 16 to 150. At mT5-base layers 5–11 one direction holds all the variance to three decimals. ABTT with ten components, fit on training vectors, removes the concentration on held-out passages too: at the collapsed layers, test top-PC share falls to 0.03–0.05 and effective rank rises to 148–173.

Correlations with ranking. Across all 100 panel layers, top-PC share and effective rank reach Spearman -0.77 and $+0.77$ with baseline AUROC (Table 3). Both values are inflated by the two clusters of collapsed and healthy layers, and layers within a model are not independent. Within the 26 collapsed layers, top-PC share carries no ordering (Spearman $+0.24$, $p = 0.23$). Within the 74 healthy layers it orders them only moderately (-0.49). Mean pairwise cosine correlates positively with AUROC within the three panel T5 encoders ($+0.67$ on test passages, $+0.72$ on training passages). This pattern does not carry over to T5-v1.1-base, whose collapsed layers sit at mean cosines of 0.87 to 0.96 on training passages, around the 0.91 and 0.92 of its two healthy layers; within that model the correlation is -0.64. The raw cosine gap, the margin a routing threshold works with, tracks ranking poorly: across the panel it correlates with AUROC at Spearman $+0.22$. The absence of a reliable ordering agrees with reports that geometric statistics do not reliably predict downstream performance (Kulkarni et al., 2026; Nastase and Merlo, 2025).

Cosine spread at the collapsed layers. At the collapsed layers, test cosines spread across both signs. At mT5-base's collapsed layers the 5th percentile lies between -0.83 and -0.77 and the median between 0.84 and 0.90, with a standard deviation of about 0.75. At LaTa's collapsed layers the 5th percentile lies between -0.50 and -0.26, and the standard deviation reaches 0.63. In the embedding-trained models the standard deviation stays between 0.007 and 0.10 at every layer. Pooled passages at a collapsed layer therefore lie on both sides of one dominant direction, which Section 6 examines.

SIF at the collapsed layers. At the collapsed layers SIF recovers about 80 percent of the ABTT gain in LaTa and PhilTa, and under 10 percent in mT5-base (Table 4). It stays below ABTT at every collapsed layer.

Score distributions. Figure 4 restates the collapse and the repair as score distributions rather than as summary metrics. At PhilTa's collapsed layer 8 the cosines of same-source and different-source pairs overlap almost completely, and ABTT separates them. Appendix J maps the whole collection for the three raw T5-family panel models.

	All (100)		**Collapsed** (26)		**Healthy** (74)		**LOMO** ρ	
Statistic	r	ρ	r	ρ	r	ρ	min	max
Mean pairwise cosine	0.83	0.55	0.05	0.49	−0.19	−0.05	0.42	0.67
Top-PC share	−0.92	−0.77	0.40	0.24	−0.51	−0.49	−0.83	−0.70
Effective rank	0.80	0.77	−0.18	−0.23	0.53	0.48	0.70	0.85
Raw cosine gap	0.24	0.22	0.98	0.92	0.24	0.12	−0.10	0.48
Standardized gap	0.86	0.97	0.95	0.92	0.90	0.93	0.96	0.98

Table 3: Pearson (r) and Spearman (ρ) correlation of per-layer statistics with baseline test ranking AUROC, over the 100 panel layers (All), the 26 collapsed layers (baseline AUROC below 0.70) and the 74 healthy ones. LOMO: range of ρ over all layers with one model left out. Mean pairwise cosine, top-PC share and effective rank are label-free; top-PC share and effective rank use training passages, the rest test passages. Raw cosine gap: mean same-source minus mean different-source cosine; standardized gap: raw gap over the standard deviation of test cosines. Within the collapsed layers top-PC share has $\rho = +0.24$ ($p = 0.23$); layers within a model are not independent, so the p-values overstate the evidence.

Model	n	Base	SIF	ABTT	SIF share (%)
LaTa	10	0.502 0.496–0.563	0.866 0.842–0.938	0.970 0.962–0.973	78.3
PhilTa	9	0.542 0.538–0.601	0.891 0.872–0.928	0.981 0.973–0.984	79.8
mT5-base	7	0.657 0.654–0.660	0.681 0.672–0.689	0.978 0.974–0.982	7.7

Table 4: Test ranking AUROC over the collapsed layers (n; baseline AUROC below 0.70) of LaTa, PhilTa and mT5-base: median, with the range below. SIF only reweights tokens before mean pooling and removes no direction; ABTT (D chosen per layer on the training split) is the per-layer record of Table 17. SIF share: median over the collapsed layers of the percentage of the Base-to-ABTT gain that SIF recovers.

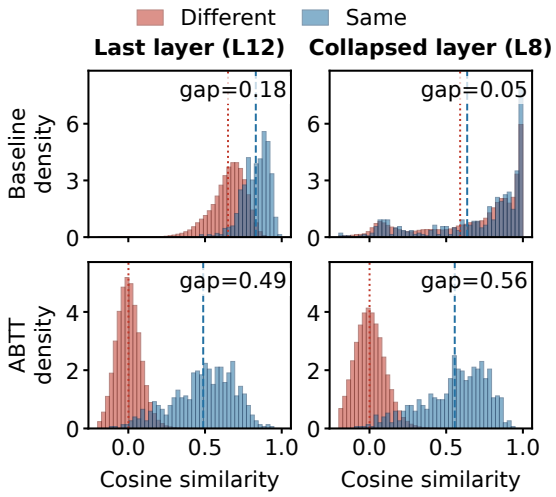


Figure 4: Test cosine distributions for PhilTa's last layer and collapsed layer 8, before and after ABTT. Same: pairs sharing a directory; Different: the rest; dashed, dotted lines: Same, Different means; gap: their difference. At layer 8 the two overlap almost completely until ABTT separates them.

D All Thirteen Models

This appendix supports Section 5. Table 5 extends Table 1 in two ways. It adds two embedding-trained models that Table 1 leaves out, Sentence-T5 (Ni et al., 2022) and SPhilBERTa (Riemenschneider and Frank, 2023b). It also gives each checkpoint's feed-forward type and width, whether its embeddings are tied, its lowest effective rank and its range of mean pairwise cosine. None of the added models collapses. Every checkpoint has at most 0.6B parameters, and each is released under the Apache 2.0 license, as its Hugging Face model card states. The rest of the appendix gives the readouts behind the sibling pairs, T5-base against T5-v1.1-base and T5-efficient-base, the projection on the controls, the English-text check and the fine-tuned LaTa model. Per-layer values are in the code release.

The sibling pairs. Table 6 gives the siblings by layer: LaBERTa rises from 0.826 at layer 1 to 0.910 at layer 12, and PhilBERTa stays between 0.883 and 0.923. Effective rank falls to 1.36 in LaTa and 1.81 in PhilTa, and never below 57.87 in LaBERTa or 82.12 in PhilBERTa (Table 5). Within each pair, several factors differ together. The T5 member is the encoder of an encoder-decoder, pretrained with span corruption; the sibling is an encoder-only RoBERTa-style model, pretrained with masked language modeling (Riemenschneider and Frank, 2023a). The T5 members use a SentencePiece tokenizer and the siblings a byte-level BPE tokenizer, so the tokens that enter the mean also differ. By their checkpoint configurations, the T5 members also use relative position biases, a gated-GELU feed-forward block and untied embeddings, and the siblings learned absolute positions, a GELU feed-forward block and tied embeddings. Two input details of the siblings' tokenizers matter for Table 6. LaBERTa's tokenizer encodes every space as its own token, so the indentation in the corpus text becomes tokens, which the pooling filter drops from the mean. PhilBERTa's published tokenizer adds no sequence-boundary tokens, so the two whitespace-only witnesses pool to zero vectors; scoring without those two witnesses changes no AUROC by more than 0.0003.

T5-base and T5-v1.1-base. The two checkpoints share their tokenizer, which splits the Latin corpus identically, the span-corruption objective, relative position biases and pre-layer RMS normalization, at the same depth and width (Raffel et al., 2020; Google Research, 2020). They differ in feed-forward block (ReLU with inner width 3072, against gated GELU with 2048), in embedding tying, and in pretraining. T5-base was pretrained on C4 mixed with supervised tasks, with dropout (Raffel et al., 2020); T5-v1.1-base on C4 alone, without dropout by its release notes (Google Research, 2020), although its released pretraining configuration sets a dropout rate of 0.1. Dropout does not separate the collapsing models from the others, because PhilTa was also pretrained with dropout (Riemenschneider and Frank, 2023a). T5-base shows a weak form of the same depth profile: its top-PC share rises to 0.545 and its effective rank falls to 14.85 at layer 8, while its AUROC stays between 0.816 and 0.864. T5-v1.1-base recovers only partly at its last layer, to 0.772. T5-efficient-base (Tay et al., 2022) has T5-base's ReLU feed-forward with inner width 3072, tied embeddings, vocabulary, depth and width, and T5-v1.1-base's C4-only pretraining data; it was pretrained without dropout and on a much shorter run: 524,288 steps of 65,536 tokens, about 1/30 of the tokens of either, by the released training configurations. It ranks below T5-base at every layer, by 0.016 to 0.080 (Table 6), and its layer 11 has effective rank 8.1, a mild form of the low-rank profile. Its mean pairwise cosine is 0.97 to 0.99 at layers 1 to 11 and 0.89 at layer 12 (Table 5), so a high mean cosine also flags a model that does not collapse.

The projection on the controls. Table 6 applies ABTT with the panel's settings to the four controls and T5-efficient-base. Nine of their 60 layers stay below the panel's band (0.962 to 0.987): layers 6 to 12 of T5-v1.1-base (0.927 to 0.959) and layer 11 of T5-base and of T5-efficient-base (0.959 each). $D = 10$, the top of the selection grid, is selected at 46 of the 48 control layers, so a larger D might narrow the gap; we did not test this.

English text. Figure 5 gives the label-free check of Section 5. The English sample comes from the public-domain US court opinions of the Caselaw Access Project (Harvard Law School Library Innovation Lab, 2024), in the redistribution of the Common Pile (Kandpal et al., 2025); every sampled document is marked public domain. Legal prose is the nearest openly licensed English register to canon law. Each non-empty Latin passage receives one English passage from a distinct opinion, and the two empty Latin files receive an empty one. The English passage is a run of whole words from a

			Checkpoint			Over layers				Layers	
Model	Type	Emb. obj.	Feed-forward	d_{ff}	Tied	$\text{AUROC}_{\min}$	$\text{PC1}_{\max}$	$\text{Rank}_{\min}$	Mean cos.	Coll.	PC1≥0.76
Main text											
LaTa	T5	none	gated GELU	2048	no	0.496 (6)	0.952 (4)	1.36 (8)	0.22–0.58	2–11	2–11
PhilTa	T5	none	gated GELU	2048	no	0.538 (10)	0.858 (6)	1.81 (6)	0.51–0.71	3–11	3–11
mT5-base	T5	none	gated GELU	2048	no	0.654 (5)	1.000 (5)	1.00 (5)	0.33–0.95	5–11	4–11
T5-v1.1-base	T5	none	gated GELU	2048	no	0.489 (2)	0.975 (9)	1.14 (9)	0.87–0.96	2–11	2–11
T5-base	T5	none	ReLU	3072	yes	0.816 (11)	0.545 (8)	14.85 (8)	0.90–0.95	none	none
T5-efficient-base	T5	none	ReLU	3072	yes	0.736 (11)	0.627 (11)	8.10 (11)	0.89–0.99	none	none
LaBERTa	Enc.	none	GELU	3072	yes	0.826 (1)	0.254 (9)	57.87 (9)	0.70–0.91	none	none
PhilBERTa	Enc.	none	GELU	3072	yes	0.883 (6)	0.162 (6)	82.12 (11)	0.84–0.96	none	none
LaBSE	Enc.	contrastive	GELU	3072	n/a	0.806 (1)	0.524 (8)	16.19 (8)	0.73–0.93	none	none
Qwen3-0.6B	Dec.	contrastive	gated SiLU	3072	yes	0.858 (1)	0.183 (16)	71.60 (16)	0.84–0.97	none	none
KaLM-mini	Dec.	contrastive	gated SiLU	4864	yes	0.861 (5)	0.419 (5)	23.83 (5)	0.59–0.93	none	none
Appendix only											
Sentence-T5	T5	contrastive	ReLU	3072	n/a	0.800 (11)	0.943 (6)	1.56 (6)	0.72–0.98	none	2–9
SPhilBERTa	Enc.	distillation	GELU	3072	n/a	0.892 (1)	0.127 (11)	64.67 (11)	0.61–0.95	none	none

Table 5: All thirteen models, read on the Latin corpus as mean-pooled hidden states with no correction; Type, $\text{AUROC}_{\min}$ and $\text{PC1}_{\max}$ as in Table 1, each extreme followed by its layer. Emb. obj.: embedding objective trained after pretraining, if any. Checkpoint columns follow each configuration file (T5-base, T5-efficient-base, LaBERTa and PhilBERTa leave tying at the library default, tied; n/a: no output head). $\text{Rank}_{\min}$ and Mean cos.: lowest entropy effective rank and range of mean pairwise cosine, on training embeddings. Layers: the layers collapsed or with top-PC share of at least 0.76. Checkpoints: LaTa, `bowphs/LaTa`; PhilTa, `bowphs/PhilTa`; mT5-base, `google/mt5-base`; T5-v1.1-base, `google/t5-v1_1-base`; T5-base, `google-t5/t5-base`; T5-efficient-base, `google/t5-efficient-base`; LaBERTa, `bowphs/LaBerta`; PhilBERTa, `bowphs/PhilBerta`; LaBSE, `sentence-transformers/LaBSE`; Qwen3-0.6B, `Qwen/Qwen3-Embedding-0.6B`; KaLM-mini, `KaLM-Embedding/KaLM-embedding-multilingual-mini-instruct-v2.5`; Sentence-T5, `sentence-transformers/sentence-t5-base`; SPhilBERTa, `bowphs/SPhilBerta`.

	LaBERTa		PhilBERTa		T5-base		T5-v1.1-base		T5-efficient-base	
Layer	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT
1	0.826	0.974	0.893	0.976	0.855	0.981	0.848	0.970	0.820	0.975
2	0.832	0.977	0.905	0.977	0.864	0.981	0.489	0.969	0.822	0.968
3	0.839	0.972	0.913	0.978	0.864	0.974	0.493	0.965	0.812	0.973
4	0.830	0.969	0.903	0.974	0.849	0.971	0.505	0.963	0.798	0.974
5	0.828	0.967	0.885	0.976	0.848	0.969	0.515	0.963	0.794	0.971
6	0.834	0.963	0.883	0.974	0.843	0.967	0.523	0.959	0.803	0.967
7	0.836	0.964	0.896	0.974	0.849	0.962	0.533	0.952	0.809	0.969
8	0.844	0.963	0.905	0.975	0.847	0.966	0.536	0.945	0.812	0.970
9	0.857	0.965	0.911	0.979	0.836	0.969	0.533	0.934	0.794	0.964
10	0.873	0.968	0.919	0.978	0.826	0.962	0.531	0.930	0.782	0.962
11	0.892	0.966	0.913	0.980	0.816	0.959	0.522	0.927	0.736	0.959
12	0.910	0.967	0.923	0.976	0.843	0.973	0.772	0.959	0.827	0.964

Table 6: Test ranking AUROC of the four controls (LaBERTa, PhilBERTa, T5-base and T5-v1.1-base) and of T5-efficient-base, an added checkpoint rather than a fifth control. Base: mean-pooled hidden states without correction. ABTT: fit on training passages, with D chosen per layer on training DirAcc@1 as for the panel. Nine of these 60 layers stay below the panel's band (0.962 to 0.987): seven of T5-v1.1-base, one of T5-base and one of T5-efficient-base.

random sentence start in the body text, with an mT5 token count within $\max(2, 3\%)$ tokens of the Latin passage's, and it inherits its partner's split. The two texts then have the same mT5 length distribution (median 98 tokens). The match holds only under the mT5 tokenizer: under PhilTa's tokenizer the English passages are about 40 percent longer, and under T5-v1.1's English-only tokenizer the Latin passages are about 55 percent longer. Re-pairing the training passages on each model's own token lengths (653 to 847 pairs) moves top-PC share by at most 0.010 at the layers where it is at least 0.76. The two texts differ in language, era and genre together, and the English sample covers one register.

Fine-tuned LaTa. Table 7 gives fine-tuned LaTa at every layer, with no post-hoc correction, beside the pretrained model. At fine-tuned layers 2 to 11, top-PC share stays at 0.759 to 0.945, against 0.771

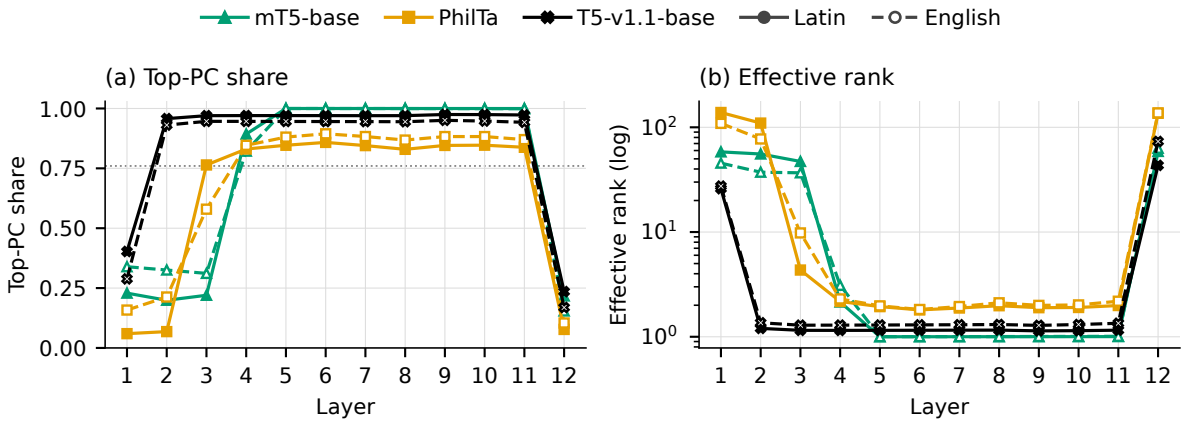


Figure 5: Label-free geometry of three T5 encoders on the Latin corpus and on English court opinions matched in mT5 token length: (a) top-PC share, dotted line at 0.76; (b) effective rank, log scale. On both texts every model is low-rank from layer 2 to 4 through layer 11 and recovers at layer 12. Top-PC share is at least 0.76 at layers 4–11 of mT5-base (Latin 0.893–1.000, English 0.822–1.000), at layers 3–11 of PhilTa on Latin (0.764–0.858) and 4–11 on English (0.846–0.894), and at layers 2–11 of T5-v1.1-base (Latin 0.958–0.975, English 0.931–0.950).

	AUROC		Top-PC share		Eff. rank	
Layer	Before	After	Before	After	Before	After
1	0.934	0.936	0.050	0.050	134.33	134.44
2	0.563	0.569	0.771	0.759	4.61	4.92
3	0.513	0.512	0.935	0.929	1.59	1.64
4	0.500	0.499	0.952	0.945	1.38	1.43
5	0.496	0.500	0.943	0.931	1.41	1.48
6	0.496	0.500	0.934	0.921	1.44	1.51
7	0.498	0.501	0.937	0.923	1.43	1.50
8	0.502	0.505	0.950	0.940	1.36	1.42
9	0.503	0.507	0.943	0.931	1.40	1.48
10	0.505	0.510	0.942	0.930	1.43	1.50
11	0.508	0.512	0.939	0.927	1.47	1.55
12	0.938	0.984	0.081	0.069	132.22	137.68

Table 7: LaTa before and after contrastive fine-tuning on training pairs, with no post-hoc correction. AUROC: test ranking AUROC. Top-PC share and effective rank are read on the training passages, which include those of the fine-tuning pairs. Fine-tuning lifts the last layer and leaves the middle layers collapsed.

to 0.952 before fine-tuning, so the collapsed geometry stays in place (fine-tuned layer 2, at 0.759, sits just below the 0.76 screen). The fine-tuning runs at most eight epochs on 499 pairs (Appendix G). It is therefore a weak test of whether an embedding objective repairs the middle layers: small-scale supervision at the output may leave them untouched whatever produces the collapse.

E Localizing the Collapse: Details

This appendix gives the per-layer tables and the further controls behind Section 6. Every ranking, component and threshold is fit on training passages, and a layer counts as restored at test AUROC 0.90.

Coordinate ablation. Table 8 reads each model at its lowest-AUROC baseline layer. The two coordinate rankings pick nearly the same coordinates at the collapsed layers: on average 9.1 of the top ten are shared. Massive coordinates alone do not mark a collapse. At LaTa layer 6, the lowest-AUROC layer of the panel, the three pooled coordinates of largest mean magnitude reach 720, 684 and 676, against a median of 8.5. At mT5-base layer 1, a healthy layer (AUROC 0.822), they reach 135, 131 and 63, against a median of 1.9. Per-coordinate standardization with training statistics (Timkey and van Schijndel, 2021) is a partial repair. It recovers a median 79% of the gain from removing ten components, but it restores only 9 of the 26 layers (1 of 10 in LaTa, 5 of 9 in PhilTa, 3 of 7 in mT5-base). The embedding-trained models barely react to zeroing: at the train-selected layers of LaBSE, Qwen3-0.6B and KaLM-mini, zeroing up to ten coordinates changes AUROC by at most 0.007.

Zeroing more coordinates. Figure 6 extends the zeroing to 400 coordinates. Under the variance ranking, zeroing restores none of the 26 layers up to 30 coordinates, 7 at 100, and 19 at 300 and 400. At the lowest-AUROC layers plotted in Figure 6, LaTa and PhilTa pass 0.90 only after 200 and 150 coordinates are zeroed (0.924 and 0.937 at $k = 400$), and mT5-base reaches only 0.779 at $k = 400$, where its top-PC share is still 0.949. The random-coordinate curves in Figures 3a and 6 are means over five random draws of the k coordinates. Zeroing 400 random coordinates removes a median 55% of the training variance, yet the median

			Zeroed, ranked by mean $\lvert x\rvert$				Zeroed, ranked by variance							
Model	L	Base	1	3	5	10	1	3	5	10	Std.	D=0	ABTT$_{D=1}$	ABTT$_{D=3}$
Test ranking AUROC														
LaTa	6	0.496	0.498	0.514	0.512	0.574	0.498	0.514	0.516	0.574	0.822	0.482	0.671	0.973
PhilTa	10	0.538	0.537	0.540	0.540	0.557	0.538	0.538	0.544	0.558	0.907	0.501	0.602	0.964
mT5-base	5	0.654	0.655	0.660	0.694	0.716	0.655	0.660	0.694	0.716	0.806	0.670	0.835	0.914
LaBSE	1	0.806	0.804	0.793	0.794	0.788	0.802	0.788	0.775	0.790	0.907	0.867	0.886	0.929
Qwen3-0.6B	1	0.858	0.866	0.874	0.877	0.880	0.866	0.870	0.873	0.874	0.963	0.954	0.959	0.970
KaLM-mini	5	0.861	0.862	0.857	0.857	0.862	0.865	0.865	0.869	0.863	0.933	0.917	0.926	0.969
Top-PC share of the training vectors														
LaTa	6	0.934	0.921	0.885	0.859	0.712	0.921	0.885	0.851	0.712	0.211	0.934	0.712	0.071
PhilTa	10	0.847	0.827	0.796	0.760	0.580	0.828	0.777	0.733	0.608	0.143	0.847	0.763	0.088
mT5-base	5	1.000	1.000	1.000	0.998	0.995	1.000	1.000	0.998	0.995	0.910	1.000	0.458	0.138
LaBSE	1	0.263	0.263	0.267	0.258	0.199	0.202	0.206	0.205	0.213	0.189	0.263	0.138	0.077
Qwen3-0.6B	1	0.100	0.101	0.096	0.095	0.095	0.101	0.096	0.097	0.098	0.102	0.100	0.077	0.049
KaLM-mini	5	0.419	0.298	0.301	0.306	0.192	0.284	0.195	0.193	0.198	0.196	0.419	0.210	0.067

Table 8: Coordinate ablation at each model's lowest-AUROC baseline layer (L), fit on training embeddings. Std.: per-coordinate standardization; D=0: centering alone, which leaves top-PC share unchanged by definition. Figure 6 extends the zeroing to 400 coordinates. In the three T5-family rows, zeroing ten coordinates leaves AUROC at or below 0.716, while standardization reaches 0.806 to 0.907 and ABTT$_{D=3}$ 0.914 to 0.973.

AUROC stays at 0.538, against 0.541 uncorrected. At the layers plotted in Figure 6, random zeroing moves the AUROC of LaTa, PhilTa and mT5-base by at most 0.010, and zeroing by variance moves the embedding-trained models by at most 0.038. Top-PC share falls before AUROC recovers: at 100 zeroed coordinates it is below 0.2 at 19 layers, while only 7 are restored.

What survives zeroing. A follow-up zeroes the ten highest-variance coordinates and asks which direction then dominates. The new dominant direction is the remainder of the first component on the other coordinates (median cosine 0.999 with that remainder). Removing the original second and third components while keeping the first restores none of the 26 layers. Zeroing followed by ABTT with one, two or three components restores 9, 21 and 26 layers, against 1, 13 and 26 without zeroing.

Number of directions. Table 9 gives the sweep over D at each model's lowest-AUROC layer and as medians over groups of layers. LaTa reaches the full gain of removing ten components with two (median 98%), and PhilTa with three (95%). mT5-base reaches a median 85% with three and keeps rising up to ten. With three components every collapsed layer recovers at least 80% of that gain. The failure of a single component is not an estimation artifact. At every collapsed layer the first component fit on training passages and the one fit on test passages agree to $|\cos| \geq 0.994$. The selection grid stops at $D = 10$; extending it to $D = 50$ changes test AUROC by at most 0.023.

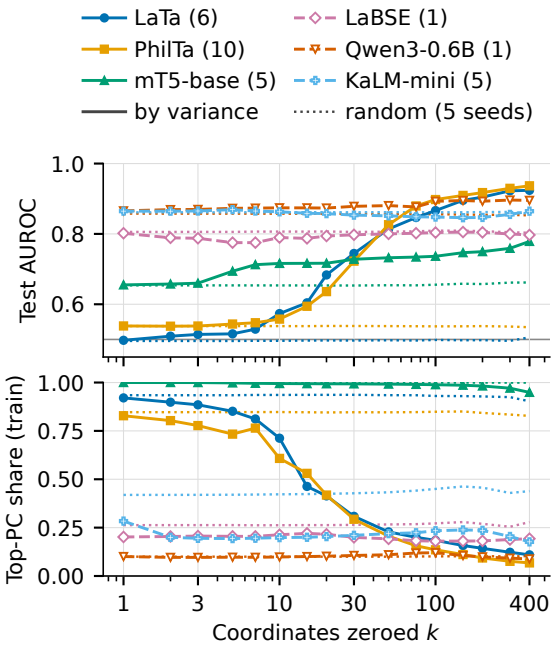


Figure 6: Test ranking AUROC (top) and training top-PC share (bottom) against zeroed coordinates k (log axis), each model at its lowest-AUROC baseline layer (in parentheses). Filled, solid (raw T5-family) and hollow, dashed (embedding-trained): zeroing by variance; dotted: random coordinates (five seeds); gray line: chance. At these layers zeroing by variance passes 0.90 for LaTa and PhilTa only past 100 coordinates, and not for mT5-base.

Removed and retained subspaces. Table 10 ranks passages within the removed and within the retained subspace. At the collapsed layers the removed subspace holds a median 99% of the centered variance of vectors that already rank at

			Removed components D										
Layers	n	raw	0	1	2	3	5	7	10	15	20	30	50
Lowest-AUROC baseline layer of each model													
LaTa (6)	1	0.496	0.482	0.671	0.940	0.973	0.973	0.977	0.964	0.973	0.966	0.981	0.986
PhilTa (10)	1	0.538	0.501	0.602	0.888	0.964	0.977	0.982	0.982	0.975	0.977	0.973	0.969
mT5-base (5)	1	0.654	0.670	0.835	0.892	0.914	0.951	0.971	0.978	0.972	0.965	0.965	0.967
LaBSE (1)	1	0.806	0.867	0.886	0.915	0.929	0.962	0.967	0.977	0.977	0.977	0.977	0.979
Qwen3-0.6B (1)	1	0.858	0.954	0.959	0.962	0.970	0.972	0.979	0.978	0.980	0.981	0.976	0.984
KaLM-mini (5)	1	0.861	0.917	0.926	0.969	0.969	0.983	0.982	0.983	0.978	0.974	0.974	0.976
Median over model-layers													
Collapsed T5	26	0.541	0.503	0.713	0.902	0.964	0.973	0.977	0.977	0.974	0.975	0.974	0.971
LaTa	10	0.502	0.488	0.713	0.956	0.973	0.972	0.976	0.970	0.971	0.969	0.980	0.983
PhilTa	9	0.542	0.502	0.602	0.872	0.963	0.980	0.981	0.981	0.975	0.978	0.973	0.970
mT5-base	7	0.657	0.678	0.865	0.894	0.934	0.955	0.974	0.978	0.975	0.974	0.965	0.962
Other T5	10	0.874	0.936	0.952	0.967	0.973	0.979	0.976	0.973	0.972	0.972	0.973	0.971
Embedding-trained	64	0.903	0.944	0.966	0.971	0.976	0.979	0.980	0.978	0.977	0.975	0.974	0.973
All	100	0.885	0.936	0.958	0.968	0.973	0.978	0.979	0.977	0.975	0.974	0.974	0.972

Table 9: Test ranking AUROC against the number D of components ABTT removes, fit on training embeddings ($D = 0$: centering alone). Top: each model at its lowest-AUROC baseline layer (in parentheses). Bottom: medians over each group's n model-layers. Three components bring every collapsed layer to 0.91 or above; mT5-base keeps improving to ten.

chance, so its chance-level ranking there is close to a tautology. The next ten components alone rank at a median 0.919 at those layers, a dimension-matched control (Table 10). The healthy layers are more informative. At the 74 healthy panel layers the removed subspace still ranks at 0.736 to 0.937, so ABTT discards some signal. Yet at all 36 layers of the three pretrained T5 encoders the removed directions rank below the next D components. At LaTa's last layer, contrastive fine-tuning moves the top ten directions from below the next ten (0.868 against 0.907) to above them (0.953 against 0.900), consistent with Rajaee and Pilehvar (2021).

Token audit. Table 11 gives the token audit at each model's lowest-AUROC layer. Its pooling columns compare mean pooling with four variants: without special tokens, SIF weights with special tokens kept, SIF with training-only token frequencies (which also drops special tokens), and without the 100 most frequent training tokens. The ablation column pools without the m token types that contribute most to the top principal components, for m from 1 to 100 under either of two token rankings. Its control is the mean over five random sets of other token types, sampled in proportion to training count, that hold at least as many training tokens as the dropped types. The token-mix columns compare against 20 random directions orthogonal to the top ten principal components. The group shares split the variance of the first component's score by covariance among special tokens, the 100 most frequent tokens and all other tokens, and $|\rho|$ is the absolute Spearman correlation of that score with log token count. In LaTa, a token mix with one mean contribution per token type, fit on training tokens, explains a median 0.985 of the variance of the first component's score on test passages, against 0.18 for random directions. In PhilTa it explains a median 0.42, and the ablation needs more types: dropping the top 3, 10 and 30 restores 1, 8 and all 9 of its collapsed layers on test passages. With the ranking and m chosen per layer on training AUROC, every collapsed layer of both models is restored (test medians: LaTa 0.926, PhilTa 0.928; minima 0.921 and 0.908). In mT5-base the token mix explains a median 0.066, against 0.160 for random directions, and dropping its top 100 types lifts AUROC only to 0.724 to 0.748.

				Full vector		Removed subspace			Retained subspace		
Model	ℓ	D	Var.	Raw	Cent.	PC1	PCs 2–D	PCs 1–D	Next D	Rand. D	All
Pretrained, train-selected layer and D											
LaTa	12	10	0.35	0.938	0.972	0.670	0.825	0.868	0.907	0.861	0.971
PhilTa	9	10	0.99	0.539	0.501	0.462	0.545	0.487	0.922	0.873	0.983
mT5-base	2	10	0.57	0.822	0.910	0.610	0.735	0.751	0.896	0.866	0.975
LaBSE	11	10	0.48	0.960	0.958	0.647	0.932	0.909	0.958	0.902	0.987
Qwen3-0.6B	2	10	0.44	0.863	0.940	0.607	0.792	0.798	0.906	0.882	0.973
KaLM-mini	1	10	0.41	0.887	0.958	0.518	0.846	0.819	0.910	0.880	0.981
Collapsed T5, median (26)	–	–	0.99	0.541	0.503	0.463	0.601	0.489	0.919	0.874	0.977
Fine-tuning contrast, same layer, $D = 10$											
LaTa, pretrained	12	10	0.35	0.938	0.972	0.670	0.825	0.868	0.907	0.861	0.971
LaTa, fine-tuned	12	10	0.33	0.984	0.985	0.697	0.935	0.953	0.900	0.874	0.970
Qwen3-0.6B, pretrained	27	10	0.41	0.967	0.969	0.643	0.913	0.917	0.909	0.884	0.982
Qwen3-0.6B, fine-tuned	27	10	0.31	0.993	0.996	0.684	0.980	0.983	0.969	0.927	0.992
KaLM-mini, pretrained	24	10	0.32	0.956	0.979	0.663	0.909	0.924	0.909	0.855	0.981
KaLM-mini, fine-tuned	24	10	0.23	0.997	0.996	0.699	0.977	0.978	0.967	0.918	0.994

Table 10: Test ranking AUROC of cosine within parts of the pooled vector, with components fit on training embeddings. Var.: share of centered training variance removed; PC1: first component alone (cosine is a sign match); Next D: PCs $D{+}1$ to $2D$; Rand. D: random D-dimensional projections of the retained vectors (mean of five seeds). At collapsed layers the signal lies outside the removed directions.

		AUROC by pooling					Token ablation		Token-mix EV		Share by token group			
Model	L	Mean	No spec.	SIF+spec.	SIF	No freq.	Drop	Rand.	PC1	Rand.	Spec.	Freq.	Other	$\lvert\rho\rvert$
LaTa	6	0.496	0.493	0.860	0.878	0.904	0.922	0.483	0.98	0.19	0.01	0.97	0.02	0.38
PhilTa	10	0.538	0.496	0.677	0.879	0.692	0.917	0.540	0.44	0.14	0.23	0.75	0.02	0.11
mT5-base	5	0.654	0.653	0.675	0.673	0.726	0.724	0.656	0.07	0.24	0.00	0.53	0.47	0.20
LaBSE	1	0.806	0.809	0.812	0.818	0.816	0.827	0.800	0.36	0.73	0.09	0.37	0.53	0.95
Qwen3-0.6B	1	0.858	0.855	0.856	0.852	0.869	0.885	0.836	0.99	0.71	0.74	0.08	0.18	0.58
KaLM-mini	5	0.861	0.859	0.867	0.863	0.872	0.879	0.854	0.98	0.47	0.01	0.02	0.97	0.51

Table 11: Token audit (test ranking AUROC) at each model's lowest-AUROC baseline layer (L). No spec.: no special tokens; SIF+spec.: SIF keeping special tokens; No freq.: no 100 most frequent tokens; Drop: no top m token types, m and ranking chosen on training AUROC; Rand.: mass-matched random types. Token-mix EV: PC1 score variance explained by token mix; Rand.: random directions. Group shares split it and sum to one, but can fall outside [0, 1]. $\lvert\rho\rvert$: Spearman correlation with log token count.

F Bootstrap Intervals and Routing Checks

This appendix gives the intervals behind Table 2 and the routing checks that Section 7 quotes.

Intervals. A bootstrap replicate draws the 514 test directories with replacement ($B = 10{,}000$, seed 233). A drawn directory keeps every file and pair: a directory drawn c times weights its files and within-directory pairs by c and its pairs with a directory drawn c' times by cc', and no pair is formed between two copies of one directory. The layer, threshold, ABTT components, D, and SIF weights stay fit on the training split, and every score is computed once on the full test set, so a replicate only reweights test pairs and files. Intervals are 95% percentiles; a paired difference recomputes both cells on each replicate. Before resampling, every cell was recomputed with the paper's evaluation code and matched its printed value. Table 12 gives every cell and Table 13 the contrasts quoted in Section 7.

Routing checks. Table 14 reports, at the headline routing cells of the three embedding-trained encoders, the AUROC of each test witness's maximum cosine for existing against new witnesses (no threshold), the gain in assignment accuracy from the best test threshold over the training one, the skewness of the 10-occurrence distribution (Radovanović et al., 2010), centering alone ($D = 0$), and assignment accuracy with τ refit as the exact best-F1 cut over all training pair scores.

Routing gain without a threshold. For the three embedding-trained encoders, we compare the ABTT cell with the baseline cell, each at its own train-selected routing layer, with paired 95% intervals. The existing-versus-new AUROC of each test witness's best-match cosine rises by 0.037 (0.021 to 0.055) for LaBSE, 0.055 (0.034 to 0.077) for Qwen3-0.6B, and 0.027 (0.012 to 0.043) for KaLM-

mini. At the baseline cells the train-fit threshold comes within 1.0 point of the best test threshold. When both cells use their best test threshold, ABTT still raises assignment accuracy by 7.0 points (4.2 to 9.3) for LaBSE, 8.9 (6.1 to 11.7) for Qwen3-0.6B, and 4.4 (2.2 to 6.8) for KaLM-mini. From the baseline cell to the ABTT cell, the skewness of the 10-occurrence distribution falls from 2.0–2.5 to 0.6–1.1 (Table 14). We report this drop in hubness as an observation and do not test it as a cause.

Threshold grid. Every printed number uses a grid of 200 evenly spaced thresholds in $[0, 1]$. With the layer and D fixed, a grid of 10,001 values moves 45 of the 66 printed routing numbers, by a median of 3 files (0.35 points) and at most 2.6 points (mT5-base's baseline). The exact best-F1 cut over training pairs moves 48. Re-selecting the layer and D as well moves 42 (fine grid) and 44 (exact cut) of the 60 encoder routing numbers, and no conclusion of Section 7 changes.

		Ranking	Routing	
Model	**Setting**	AUROC	Assignment acc.	DirAcc@1
LaTa	Base	0.938 [0.910, 0.963]	73.8 [70.2, 77.3]	72.1 [68.4, 75.8]
	SIF	0.948 [0.915, 0.974]	83.2 [80.1, 86.3]	81.8 [78.5, 85.1]
	ABTT	0.971 [0.948, 0.991]	88.5 [85.8, 90.9]	86.1 [83.2, 88.9]
	SIF+ABTT	0.966 [0.938, 0.989]	90.2 [87.6, 92.6]	88.2 [85.4, 90.9]
PhilTa	Base	0.939 [0.918, 0.958]	70.9 [66.9, 74.9]	69.3 [65.2, 73.5]
	SIF	0.955 [0.935, 0.972]	79.0 [75.6, 82.4]	77.7 [74.2, 81.3]
	ABTT	0.983 [0.969, 0.993]	91.1 [88.9, 93.2]	88.3 [85.6, 90.9]
	SIF+ABTT	0.982 [0.969, 0.993]	91.4 [89.1, 93.6]	88.9 [86.2, 91.5]
mT5-base	Base	0.838 [0.810, 0.866]	47.2 [43.1, 51.6]	46.6 [42.5, 51.0]
	SIF	0.845 [0.816, 0.876]	54.5 [50.3, 58.8]	54.1 [49.8, 58.4]
	ABTT	0.975 [0.954, 0.992]	90.3 [87.9, 92.7]	88.5 [85.7, 91.2]
	SIF+ABTT	0.974 [0.950, 0.993]	90.3 [88.0, 92.6]	87.5 [84.8, 90.2]
LaBSE	Base	0.956 [0.932, 0.977]	83.3 [80.2, 86.4]	81.5 [78.1, 84.7]
	SIF	0.956 [0.931, 0.977]	83.4 [80.2, 86.5]	81.9 [78.6, 85.2]
	ABTT	0.987 [0.977, 0.995]	90.8 [88.5, 92.9]	88.5 [85.8, 90.9]
	SIF+ABTT	0.976 [0.960, 0.989]	89.6 [87.2, 91.9]	86.9 [84.2, 89.5]
Qwen3-0.6B	Base	0.966 [0.950, 0.981]	82.3 [79.1, 85.4]	80.3 [76.9, 83.6]
	SIF	0.942 [0.919, 0.963]	76.0 [72.5, 79.4]	74.2 [70.6, 77.8]
	ABTT	0.973 [0.948, 0.990]	91.5 [89.1, 93.7]	89.4 [86.7, 92.0]
	SIF+ABTT	0.972 [0.947, 0.991]	90.8 [88.3, 93.1]	88.6 [85.8, 91.2]
KaLM-mini	Base	0.972 [0.955, 0.986]	87.5 [84.8, 90.1]	85.9 [82.9, 88.7]
	SIF	0.964 [0.944, 0.982]	85.7 [82.8, 88.5]	84.0 [80.9, 87.1]
	ABTT	0.981 [0.963, 0.993]	91.7 [89.5, 93.8]	89.4 [86.7, 91.9]
	SIF+ABTT	0.981 [0.962, 0.995]	90.8 [88.4, 93.0]	88.0 [85.2, 90.7]
LaTa (fine-tuned)	Base	0.984 [0.973, 0.992]	83.4 [80.2, 86.5]	81.6 [78.1, 84.9]
	ABTT	0.970 [0.947, 0.990]	87.8 [85.2, 90.2]	85.2 [82.3, 87.9]
Qwen3-0.6B (fine-tuned)	Base	0.996 [0.993, 0.999]	91.4 [88.7, 93.8]	90.8 [88.1, 93.3]
	ABTT	0.994 [0.989, 0.998]	92.1 [89.6, 94.3]	91.4 [88.8, 93.7]
KaLM-mini (fine-tuned)	Base	0.997 [0.993, 0.999]	92.4 [90.1, 94.6]	91.7 [89.3, 94.0]
	ABTT	0.994 [0.988, 0.998]	93.4 [91.2, 95.4]	92.5 [90.3, 94.7]
BM25 (word)	–	0.974 [0.956, 0.987]	62.5 [58.2, 66.6]	56.3 [52.0, 60.5]
Char. n-grams	–	0.987 [0.976, 0.996]	91.8 [89.7, 93.8]	89.9 [87.3, 92.3]
Levenshtein	–	0.952 [0.927, 0.975]	87.8 [85.0, 90.4]	86.2 [83.2, 89.2]

Table 12: Every headline cell of Table 2, plus SIF and SIF+ABTT, at its train-selected layer, with intervals in brackets. The last block gives the three lexical reference systems: BM25 over words, TF-IDF over character 3–5-grams (Char. n-grams), and Levenshtein similarity. Routing in percent. Directory bootstrap, 10,000 replicates, 95% percentile intervals; protocol in Appendix F.

Contrast	**Model**	Δ AUROC	Δ Assignment acc.	Δ DirAcc@1
ABTT − Base	LaTa	+0.034 [0.017, 0.054]	+14.7 [11.1, 18.3]	+14.0 [10.4, 17.6]
	PhilTa	+0.043 [0.029, 0.060]	+20.3 [16.2, 24.5]	+19.0 [15.0, 23.1]
	mT5-base	+0.137 [0.109, 0.166]	+43.1 [38.6, 47.6]	+41.8 [37.2, 46.4]
	LaBSE	+0.030 [0.014, 0.049]	+7.5 [4.6, 10.4]	+7.0 [4.1, 10.0]
	Qwen3-0.6B	+0.006 [−0.010, 0.019]	+9.2 [6.3, 12.2]	+9.1 [6.1, 12.2]
	KaLM-mini	+0.009 [−0.002, 0.021]	+4.2 [1.7, 6.7]	+3.5 [1.0, 6.0]
Fine-tuned: ABTT − Base	LaTa	−0.014 [−0.032, 0.001]	+4.3 [1.7, 7.1]	+3.6 [1.0, 6.3]
	Qwen3-0.6B	−0.002 [−0.005, 0.001]	+0.7 [−0.7, 2.1]	+0.6 [−0.8, 2.0]
	KaLM-mini	−0.003 [−0.007, 0.001]	+0.9 [−0.3, 2.2]	+0.8 [−0.5, 2.1]
Fine-tuned ABTT − zero-shot ABTT	LaTa	−0.002 [−0.010, 0.005]	−0.7 [−2.6, 1.2]	−0.9 [−2.8, 1.0]
	Qwen3-0.6B	+0.022 [0.006, 0.044]	+0.6 [−1.6, 2.9]	+2.0 [0.0, 4.0]
	KaLM-mini	+0.013 [0.002, 0.029]	+1.6 [−0.3, 3.6]	+3.1 [1.2, 5.2]
Char. n-grams − ABTT	LaTa	+0.016 [0.003, 0.030]	+3.4 [1.1, 5.7]	+3.7 [1.4, 6.2]
	PhilTa	+0.004 [−0.002, 0.011]	+0.7 [−1.0, 2.4]	+1.5 [−0.2, 3.3]
	mT5-base	+0.012 [0.000, 0.027]	+1.5 [−0.6, 3.7]	+1.4 [−0.7, 3.6]
	LaBSE	0.000 [−0.004, 0.005]	+1.0 [−1.0, 3.1]	+1.4 [−0.8, 3.6]
	Qwen3-0.6B	+0.014 [0.002, 0.031]	+0.3 [−1.6, 2.3]	+0.5 [−1.4, 2.5]
	KaLM-mini	+0.006 [−0.002, 0.016]	+0.1 [−1.6, 1.9]	+0.5 [−1.3, 2.3]
ABTT − centering (ABTT layer)	LaTa	−0.001 [−0.016, 0.012]	+40.4 [35.8, 44.8]	+49.2 [44.1, 53.9]
	PhilTa	+0.482 [0.460, 0.504]	+7.3 [4.9, 10.0]	+6.6 [4.1, 9.3]
	mT5-base	+0.065 [0.050, 0.081]	+24.5 [20.7, 28.3]	+27.7 [23.7, 31.7]
	LaBSE	+0.029 [0.013, 0.048]	+12.1 [8.8, 15.5]	+11.7 [8.4, 15.0]
	Qwen3-0.6B	+0.033 [0.018, 0.050]	+25.9 [22.0, 29.9]	+27.7 [23.5, 32.1]
	KaLM-mini	+0.023 [0.008, 0.040]	+12.0 [9.1, 15.1]	+13.8 [10.7, 17.0]
Spread over six models	Base	0.134 [0.110, 0.158]	40.3 [35.7, 44.7]	39.3 [34.6, 43.7]
	ABTT	0.015 [0.006, 0.034]	3.3 [1.8, 5.8]	3.3 [1.8, 6.0]

Table 13: Paired differences between headline cells at train-selected layers, with intervals in brackets. Centering rows use each column's ABTT layer. Spread: largest minus smallest of the six zero-shot models. Routing in points. Differences use unrounded cells, so they can differ by one last digit from printed-cell differences. Directory bootstrap, 10,000 replicates, 95% percentile intervals; protocol in Appendix F.

Model	**Setting**	**L**	Exist./new AUROC	Assign.	Oracle gap	Exact	Skew
LaBSE	Base	11	0.922 [0.901, 0.942]	83.3	+1.0 [0.1, 3.5]	82.5	2.00
	Centering @ ABTT L	11	0.891 [0.866, 0.914]	78.7	+2.8 [0.7, 6.3]	79.0	1.84
	ABTT	11	0.960 [0.944, 0.973]	90.8	+0.6 [0.1, 1.6]	90.8	0.58
Qwen3-0.6B	Base	28	0.900 [0.874, 0.923]	82.3	+1.0 [0.2, 2.6]	82.6	2.51
	Base @ ABTT L	5	0.708 [0.666, 0.748]	46.4	+23.4 [17.7, 30.4]	53.0	2.49
	Centering @ ABTT L	5	0.802 [0.767, 0.835]	65.6	+9.6 [5.9, 14.5]	66.6	1.77
	ABTT	5	0.954 [0.935, 0.971]	91.5	+0.7 [0.2, 1.7]	91.5	1.07
KaLM-mini	Base	23	0.931 [0.910, 0.950]	87.5	0.0 [0.0, 1.1]	86.2	1.97
	Base @ ABTT L	3	0.756 [0.719, 0.792]	60.4	+11.3 [6.7, 17.0]	58.3	3.58
	Centering @ ABTT L	3	0.881 [0.853, 0.906]	79.7	+2.7 [1.2, 4.8]	79.5	1.75
	ABTT	3	0.958 [0.941, 0.973]	91.7	+0.2 [0.0, 1.6]	91.7	0.87

Table 14: Routing checks at the headline routing cells of the three embedding-trained encoders: Base and ABTT at their train-selected layers (L), baseline and centering ($D = 0$) at the ABTT layer. Exist./new AUROC: threshold-free, from each witness's maximum cosine. Assign.: assignment accuracy (%) at the threshold τ fit on training pairs. Oracle gap: accuracy at the best test threshold minus Assign. Exact: Assign. with τ at the exact training best-F1 cut. Skew: skewness of the 10-occurrence distribution (Radovanović et al., 2010). Brackets: 95% bootstrap intervals.

G Reference Systems

Table 2 carries four reference rows below the six encoders: one supervised fine-tuning ceiling for each of the three fine-tuned models, and the character 3–5-gram TF-IDF reference. This appendix describes them. Table 15 gives the three fine-tuning ceilings with the zero-shot rows they are compared against. Some routing numbers are means over *five reseedings*: each reseeding keeps the training/test split and redraws, within each directory's test witnesses, which serve as queries and which as references; queries are routed against the reference directories. Every model we fine-tuned is reported, on whichever side of its own zero-shot row it lands.

Three encoders are fine-tuned under one recipe, LaTa, Qwen3-0.6B and KaLM-mini. Fine-tuning only the Latin-pretrained model would leave the ceiling confounded with Latin pretraining, so the second model is a multilingual decoder-only encoder with no Latin-specific pretraining, Qwen3-0.6B. The third, KaLM-mini, has the best ABTT cell in the paper (91.7 assignment accuracy; 89.4 DirAcc@1, tied with Qwen3-0.6B), so it tests whether supervision can clear that cell.

All three are trained with symmetric InfoNCE over positive pairs and in-batch negatives (no two pairs from one directory in a batch), temperature 0.05, AdamW at learning rate 2×10^{-5} with weight decay 0.01, linear warmup over the first 10 percent of steps then linear decay, gradient clipping at 1.0, 16 pairs per batch, up to 8 epochs with early stopping after 3 epochs without dev improvement, and seed 42. All three are mean-pooled with the same token filter as their zero-shot rows, so the fine-tuned vectors live in the space those rows are measured in; Qwen3-0.6B and KaLM-mini add gradient checkpointing, which recomputes activations rather than storing them and leaves the gradients unchanged.

The dev slice holds out 28 of the 190 train directories with at least two files (71 files), the same for every model, so each model trains on 499 of the 565 positive train pairs. The checkpoint is chosen by dev DirAcc@1, with dev AUROC as the tie-break. Differences of one dev file or less count as ties, since one file is the resolution of a 71-file pool; GPU non-determinism alone flipped one such file between two otherwise identical Qwen3-0.6B runs. LaTa selects epoch 7, and its ABTT sweep selects $D = 10$, the top of the grid, at every layer. Qwen3-0.6B selects epoch 3 of 6 run, with $D = 2$ at its routing layer; KaLM-mini selects epoch 1 of 4 run, with the top of the grid at its routing layer. Training loss reaches 4×10^{-5} on Qwen3-0.6B and 3×10^{-5} on KaLM-mini, so the 499 pairs are memorized and each row is a ceiling at this training budget.

Compute. Most hidden-state extraction and all fine-tuning ran on a single NVIDIA A100 GPU; the control-model extraction and the token audit ran on a single NVIDIA A40, and T5-efficient-base, with a re-extraction of the four controls for Table 6, on CPU. The whole project, exploratory runs included, used about 160 GPU-hours and under 200 CPU core-hours. Every post-hoc analysis (ABTT, whitening, zeroing, bootstrap intervals) ran on CPU from cached vectors.

With ABTT, fine-tuned LaTa routes below every zero-shot ABTT cell in point estimate. It reaches 87.8 assignment accuracy and 85.2 DirAcc@1, against 88.5 to 91.7 and 86.1 to 89.4 for the zero-shot ABTT cells. Over five reseedings it ties LaTa's own zero-shot ABTT cell at 87.7 (Section 7). ABTT lowers the fine-tuned row's

System	Ranking		Routing		
	AUROC	Cosine gap	Assign. acc.	DirAcc@1	DirAcc@1 5 reseedings
LaTa (zero-shot)	0.938	0.237	73.8	72.1	73.1 ± 1.0
LaTa (zero-shot) + ABTT	0.971	0.525	88.5	86.1	87.7 ± 0.4
LaTa (fine-tuned)	0.984	0.387	83.4	81.6	83.4 ± 0.9
LaTa (fine-tuned) + ABTT	0.970	0.548	87.8	85.2	87.7 ± 0.8
Qwen3-0.6B (zero-shot)	0.966	0.024	82.3	80.3	82.4 ± 1.0
Qwen3-0.6B (zero-shot) + ABTT	0.973	0.553	91.5	89.4	90.5 ± 0.4
Qwen3-0.6B (fine-tuned)	0.996	0.684	91.4	90.8	91.7 ± 0.6
Qwen3-0.6B (fine-tuned) + ABTT	0.994	0.716	92.1	91.4	92.3 ± 0.2
KaLM-mini (zero-shot)	0.972	0.056	87.5	85.9	87.3 ± 0.7
KaLM-mini (zero-shot) + ABTT	0.981	0.568	91.7	89.4	90.9 ± 0.6
KaLM-mini (fine-tuned)	0.997	0.636	92.4	91.7	92.6 ± 0.8
KaLM-mini (fine-tuned) + ABTT	0.994	0.651	93.4	92.5	93.0 ± 0.5

Table 15: Fine-tuning ceiling for LaTa, Qwen3-0.6B and KaLM-mini on 499 training pairs; layers as in Table 16. Test scores; routing in percent; last column: mean ± s.d. over five reseedings. Fine-tuning raises ranking AUROC and ABTT lowers it in point estimate; on reseeded routing, fine-tuned ABTT ties zero-shot ABTT for LaTa and exceeds it for Qwen3-0.6B and KaLM-mini.

ranking AUROC from 0.984 to 0.970 in point estimate but widens its cosine gap from 0.387 to 0.548 (Table 15).

Lexical reference systems. The three lexical reference systems are BM25, character 3–5-gram TF-IDF and Levenshtein similarity; Table 12 gives all three on both tasks with intervals, and the character n-gram row is the reference quoted in Section 7. BM25 uses $k_1 = 1.5$ and $b = 0.75$, with document frequencies, vocabulary and average document length taken from training witnesses only. The character TF-IDF vectorizer is also fit on training witnesses only, and normalized Levenshtein similarity has nothing to fit. Every score matrix is min-max rescaled using its training block. The map is monotone, so AUROC is unchanged, but τ is fit on a fixed grid after it. That matters for BM25, whose τ would pin at the grid ceiling without rescaling, and is the identity for character TF-IDF, whose training scores already span $[0, 1]$. BM25 routes poorly because a raw BM25 score grows with query length, so a witness's maximum score tracks document length as much as relatedness, and the τ fit on the training split transfers badly. Over five reseedings, DirAcc@1 is 68.3 ± 7.1 for BM25, 91.1 ± 0.5 for character TF-IDF and 87.2 ± 0.3 for Levenshtein similarity; the runs are in the code release.

H Per-Layer Results

Tables 17 and 18 give one per-layer record for LaTa, PhilTa and mT5-base and one for LaBSE, Qwen3-0.6B and KaLM-mini: ranking AUROC and cosine gap, routing DirAcc@1, assignment accuracy, and existing and new decision accuracy, each under baseline mean pooling and under ABTT with D tuned per layer on the training split. Each table marks the ranking-selected and routing-selected layers, and Table 16 lists the train-selected layer behind every headline cell. Fixing D at ten gives the same AUROC range (0.962 to 0.987 at all 100 panel layers), so the result does not rest on how D is chosen. Per-layer values for every other setting are in the code release.

	Ranking layer				**Routing layer**			
Model	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	12	1	12	7	1	1	8	12
PhilTa	1	1	9	8	1	1	1	1
mT5-base	12	1	2	1	12	1	1	2
LaBSE	12	12	11	10	11	12	11	11
Qwen3-0.6B	26	22	2	2	28	27	5	5
KaLM-mini	23	23	1	2	23	22	3	1
LaTa (fine-tuned)	12	–	12	–	12	–	12	–
Qwen3-0.6B (fine-tuned)	28	–	27	–	28	–	27	–
KaLM-mini (fine-tuned)	24	–	24	–	24	–	24	–

Table 16: Train-selected layer behind each encoder cell of Tables 2, 12 and 15: the layer of highest training AUROC (ranking) or training DirAcc@1 (routing). Layer 1 is the first transformer block. Fine-tuned rows have no SIF setting.

		Ranking				Routing							
		AUROC		Cos. gap		DirAcc@1		Assign.		Existing dec.		New dec.	
Model	**Layer**	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT
LaTa	1	0.934	0.970	0.286	0.566	0.721	0.880	0.738	0.909	0.630	0.910	0.916	0.907
	2	0.563	0.969	0.019	0.554	0.383	0.869	0.439	0.900	0.153	0.905	0.913	0.892
	3	0.513	0.973	-0.048	0.555	0.350	0.882	0.477	0.908	0.256	0.882	0.842	0.950
	4	0.500	0.971	-0.067	0.565	0.303	0.881	0.513	0.903	0.389	0.893	0.718	0.920
	5	0.496	0.972	-0.069	0.572	0.297	0.875	0.502	0.900	0.374	0.892	0.715	0.913
	6	0.496	0.964	-0.069	0.569	0.297	0.874	0.503	0.895	0.378	0.884	0.712	0.913
	7	0.498	0.962	-0.072	0.564	0.302	0.868	0.501	0.890	0.372	0.875	0.715	0.916
	8†	0.502	0.965	-0.074	0.558	0.302	**0.861**	0.502	**0.885**	0.379	**0.884**	0.706	**0.885**
	9	0.503	0.965	-0.073	0.557	0.309	0.858	0.501	0.881	0.366	0.886	0.724	0.873
	10	0.505	0.971	-0.072	0.551	0.317	0.852	0.498	0.876	0.344	0.873	0.752	0.882
	11	0.508	0.973	-0.067	0.550	0.331	0.857	0.490	0.880	0.310	0.867	0.786	0.901
	12*	0.938	**0.971**	0.237	**0.525**	0.689	0.841	0.710	0.868	0.576	0.890	0.932	0.833
PhilTa	1†	0.939	0.977	0.231	0.574	0.693	**0.883**	0.709	**0.911**	0.591	**0.918**	0.904	**0.901**
	2	0.911	0.973	0.161	0.553	0.603	0.869	0.625	0.892	0.479	0.895	0.867	0.885
	3	0.601	0.973	0.085	0.549	0.389	0.865	0.455	0.890	0.193	0.897	0.889	0.879
	4	0.551	0.978	0.063	0.551	0.369	0.859	0.439	0.881	0.148	0.886	0.923	0.873
	5	0.542	0.979	0.053	0.551	0.334	0.854	0.457	0.876	0.252	0.879	0.796	0.873
	6	0.541	0.981	0.047	0.551	0.310	0.845	0.474	0.872	0.338	0.905	0.700	0.817
	7	0.542	0.979	0.046	0.554	0.325	0.847	0.469	0.874	0.301	0.901	0.746	0.830
	8	0.542	0.984	0.046	0.555	0.350	0.857	0.467	0.879	0.262	0.875	0.808	0.885
	9*	0.539	**0.983**	0.041	**0.540**	0.337	0.857	0.476	0.882	0.303	0.890	0.762	0.870
	10	0.538	0.982	0.040	0.531	0.344	0.833	0.480	0.865	0.299	0.908	0.780	0.793
	11	0.539	0.982	0.041	0.511	0.355	0.829	0.470	0.857	0.256	0.869	0.824	0.836
	12	0.911	0.982	0.181	0.487	0.528	0.854	0.549	0.879	0.338	0.880	0.898	0.876
mT5-base	1†	0.822	0.979	0.027	0.550	0.451	**0.885**	0.453	**0.903**	0.135	**0.877**	0.981	**0.947**
	2*	0.822	**0.975**	0.026	**0.536**	0.435	0.874	0.436	0.897	0.110	0.890	0.975	0.910
	3	0.804	0.968	0.025	0.517	0.445	0.830	0.449	0.850	0.161	0.794	0.926	0.941
	4	0.799	0.974	0.040	0.502	0.467	0.798	0.491	0.823	0.269	0.787	0.858	0.882
	5	0.654	0.978	0.219	0.482	0.352	0.756	0.512	0.790	0.422	0.746	0.659	0.864
	6	0.657	0.978	0.238	0.483	0.346	0.775	0.580	0.808	0.636	0.766	0.489	0.876
	7	0.656	0.980	0.244	0.482	0.352	0.775	0.565	0.815	0.594	0.793	0.517	0.851
	8	0.656	0.982	0.249	0.471	0.366	0.784	0.558	0.822	0.557	0.809	0.560	0.842
	9	0.660	0.982	0.235	0.475	0.394	0.787	0.508	0.831	0.364	0.841	0.746	0.814
	10	0.659	0.977	0.226	0.459	0.387	0.788	0.491	0.825	0.314	0.794	0.783	0.876
	11	0.659	0.974	0.213	0.449	0.378	0.765	0.479	0.797	0.297	0.763	0.780	0.854
	12	0.838	0.972	0.081	0.442	0.466	0.779	0.472	0.821	0.198	0.834	0.926	0.799

Table 17: Per-layer test results for LaTa, PhilTa and mT5-base, accuracies as fractions. Base: mean-pooled hidden states without correction; ABTT: fit on training embeddings, with D chosen per layer on the training split, as is the routing threshold τ. Existing dec., New dec.: accuracy of the existing-versus-new decision on existing and on new test witnesses, which Assign. pools. *, †: the ABTT train-selected ranking and routing layers (Table 16 also lists the Base layers), with that task's ABTT cells in bold.

		Ranking				Routing							
		AUROC		Cos. gap		DirAcc@1		Assign.		Existing dec.		New dec.	
Model	**Layer**	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT	Base	ABTT
LaBSE	1	0.806	0.977	0.060	0.520	0.465	0.851	0.477	0.876	0.222	0.879	0.898	0.873
	2	0.833	0.979	0.037	0.531	0.494	0.830	0.543	0.857	0.381	0.875	0.811	0.827
	3	0.853	0.978	0.062	0.543	0.500	0.848	0.514	0.879	0.277	0.886	0.907	0.867
	4	0.862	0.977	0.073	0.525	0.542	0.841	0.565	0.874	0.372	0.897	0.885	0.836
	5	0.864	0.968	0.066	0.501	0.521	0.811	0.543	0.847	0.327	0.845	0.901	0.851
	6	0.894	0.976	0.078	0.532	0.640	0.831	0.674	0.857	0.548	0.860	0.882	0.851
	7	0.887	0.975	0.088	0.524	0.561	0.810	0.584	0.841	0.398	0.849	0.892	0.830
	8	0.879	0.975	0.080	0.522	0.544	0.779	0.568	0.822	0.385	0.869	0.870	0.743
	9	0.916	0.979	0.090	0.565	0.608	0.823	0.612	0.851	0.421	0.871	0.929	0.817
	10	0.934	0.982	0.103	0.586	0.693	0.832	0.713	0.859	0.606	0.892	0.892	0.805
	11*†	0.960	**0.987**	0.061	**0.611**	0.815	**0.885**	0.833	**0.908**	0.785	**0.923**	0.913	**0.882**
	12	0.956	0.981	0.183	0.592	0.791	0.868	0.808	0.895	0.729	0.914	0.938	0.864
Qwen3-0.6B	1	0.858	0.978	0.033	0.564	0.521	0.875	0.526	0.908	0.264	0.921	0.960	0.885
	2*	0.863	**0.973**	0.024	**0.553**	0.558	0.871	0.579	0.887	0.437	0.869	0.814	0.916
	3	0.862	0.980	0.037	0.556	0.508	0.890	0.513	0.911	0.250	0.914	0.947	0.907
	4	0.860	0.986	0.036	0.547	0.559	0.874	0.576	0.901	0.411	0.936	0.848	0.842
	5†	0.861	0.981	0.030	0.545	0.450	**0.894**	0.464	**0.915**	0.181	**0.910**	0.932	**0.923**
	6	0.867	0.983	0.035	0.545	0.551	0.858	0.576	0.882	0.422	0.935	0.830	0.796
	7	0.876	0.980	0.033	0.544	0.507	0.859	0.513	0.885	0.250	0.890	0.947	0.876
	8	0.882	0.972	0.037	0.532	0.580	0.852	0.592	0.881	0.409	0.931	0.895	0.799
	9	0.881	0.977	0.034	0.532	0.587	0.852	0.607	0.882	0.460	0.901	0.851	0.851
	10	0.887	0.976	0.031	0.529	0.622	0.848	0.647	0.881	0.533	0.892	0.836	0.864
	11	0.895	0.977	0.030	0.530	0.548	0.830	0.551	0.867	0.293	0.903	0.978	0.808
	12	0.901	0.973	0.027	0.530	0.603	0.848	0.607	0.876	0.407	0.912	0.938	0.817
	13	0.907	0.974	0.026	0.523	0.618	0.851	0.625	0.881	0.432	0.908	0.944	0.836
	14	0.905	0.974	0.024	0.519	0.621	0.840	0.633	0.871	0.456	0.907	0.926	0.811
	15	0.901	0.971	0.022	0.514	0.627	0.848	0.639	0.880	0.469	0.901	0.920	0.845
	16	0.910	0.975	0.026	0.509	0.682	0.830	0.711	0.864	0.639	0.897	0.830	0.808
	17	0.916	0.979	0.032	0.521	0.641	0.840	0.652	0.873	0.480	0.886	0.935	0.851
	18	0.933	0.983	0.032	0.537	0.691	0.837	0.700	0.866	0.553	0.882	0.944	0.839
	19	0.936	0.980	0.035	0.541	0.676	0.837	0.684	0.866	0.514	0.871	0.966	0.858
	20	0.943	0.982	0.041	0.559	0.733	0.852	0.746	0.875	0.654	0.865	0.898	0.892
	21	0.953	0.982	0.047	0.562	0.752	0.881	0.761	0.907	0.647	0.912	0.950	0.898
	22	0.963	0.983	0.044	0.583	0.795	0.876	0.809	0.899	0.731	0.888	0.938	0.916
	23	0.961	0.984	0.038	0.578	0.742	0.886	0.752	0.907	0.622	0.914	0.966	0.895
	24	0.963	0.984	0.032	0.576	0.808	0.896	0.818	0.914	0.748	0.901	0.935	0.935
	25	0.961	0.983	0.027	0.578	0.718	0.886	0.725	0.904	0.574	0.893	0.975	0.923
	26	0.966	0.985	0.024	0.568	0.740	0.881	0.749	0.897	0.613	0.899	0.975	0.895
	27	0.967	0.982	0.017	0.559	0.787	0.876	0.808	0.894	0.793	0.877	0.833	0.923
	28	0.958	0.975	0.098	0.556	0.803	0.872	0.823	0.890	0.796	0.877	0.867	0.913
KaLM-mini	1*	0.887	**0.981**	0.063	**0.568**	0.577	0.902	0.585	0.921	0.391	0.921	0.907	0.920
	2	0.906	0.983	0.044	0.566	0.607	0.879	0.627	0.902	0.493	0.916	0.848	0.879
	3†	0.881	0.981	0.062	0.551	0.587	**0.894**	0.604	**0.917**	0.443	**0.923**	0.870	**0.907**
	4	0.871	0.983	0.060	0.553	0.538	0.887	0.544	0.917	0.308	0.931	0.935	0.895
	5	0.861	0.983	0.062	0.541	0.524	0.882	0.543	0.911	0.338	0.905	0.882	0.923
	6	0.872	0.983	0.063	0.529	0.490	0.876	0.502	0.904	0.256	0.895	0.910	0.920
	7	0.888	0.979	0.072	0.524	0.592	0.873	0.618	0.894	0.475	0.864	0.854	0.944
	8	0.890	0.976	0.072	0.524	0.597	0.860	0.619	0.881	0.467	0.862	0.870	0.913
	9	0.892	0.974	0.066	0.507	0.601	0.843	0.626	0.872	0.469	0.873	0.885	0.870
	10	0.893	0.971	0.067	0.494	0.584	0.844	0.613	0.873	0.449	0.875	0.885	0.870
	11	0.894	0.970	0.061	0.475	0.586	0.836	0.619	0.862	0.458	0.862	0.885	0.864
	12	0.905	0.970	0.071	0.475	0.622	0.850	0.653	0.878	0.507	0.862	0.895	0.904
	13	0.900	0.968	0.066	0.467	0.633	0.840	0.674	0.864	0.540	0.860	0.895	0.870
	14	0.904	0.971	0.063	0.471	0.636	0.829	0.688	0.854	0.593	0.836	0.845	0.885
	15	0.907	0.973	0.067	0.468	0.650	0.817	0.691	0.850	0.566	0.864	0.898	0.827
	16	0.917	0.972	0.071	0.489	0.676	0.819	0.695	0.841	0.550	0.849	0.935	0.830
	17	0.926	0.975	0.075	0.501	0.747	0.852	0.767	0.873	0.675	0.864	0.920	0.889
	18	0.945	0.977	0.060	0.519	0.777	0.865	0.796	0.883	0.697	0.875	0.960	0.898
	19	0.956	0.979	0.061	0.531	0.801	0.861	0.814	0.880	0.721	0.873	0.966	0.892
	20	0.960	0.980	0.054	0.555	0.789	0.817	0.803	0.837	0.705	0.791	0.966	0.913
	21	0.962	0.981	0.043	0.555	0.802	0.831	0.817	0.854	0.727	0.811	0.966	0.926
	22	0.971	0.985	0.050	0.555	0.857	0.875	0.875	0.895	0.850	0.879	0.916	0.923
	23	0.972	0.986	0.056	0.552	0.859	0.872	0.875	0.889	0.849	0.884	0.920	0.898
	24	0.956	0.981	0.243	0.520	0.833	0.858	0.846	0.882	0.787	0.892	0.944	0.867

Table 18: Per-layer test results for the embedding-trained models LaBSE, Qwen3-0.6B and KaLM-mini. Columns, marks and bold as in Table 17.

Pooling. Decoder-only embedding models often use last-token pooling, so we check that mean pooling, which every main-text result uses, does not handicap them. We compare the two poolings under the same ABTT pipeline, training and test split, and ABTT choices. Table 19 reports, for each model and pooling, the layer chosen on the training split by DirAcc@1, the rule behind the routing cells of Table 2, with test assignment accuracy, DirAcc@1 and cosine gap at that layer. Mean pooling routes better for each of the five models in this run, the two decoders included; mT5-base was not part of it.

Model	Pooling	Layer	D	Assignment acc.	DirAcc@1	Cosine gap
LaTa	Mean	8	10	0.885	0.861	0.558
	Last token	7	10	0.664	0.466	0.308
PhilTa	Mean	1	10	0.911	0.883	0.573
	Last token	11	10	0.589	0.406	0.205
LaBSE	Mean	11	10	0.908	0.885	0.610
	Last token	12	10	0.895	0.871	0.586
Qwen3-0.6B	Mean	5	10	0.915	0.894	0.545
	Last token	26	5	0.881	0.864	0.580
KaLM-mini	Mean	3	10	0.917	0.894	0.552
	Last token	24	10	0.875	0.847	0.483

Table 19: Mean versus last-token pooling under ABTT for five models; test scores, accuracies as fractions. D (components removed) and the layer (highest training DirAcc@1) are chosen on the training split. Mean pooling routes better for all five.

I SIF and Whitening Baselines

Section 3 compares SIF frequency weighting and reduced-rank whitening with ABTT; this appendix gives both. Table 4 summarizes SIF over the collapsed layers of LaTa, PhilTa and mT5-base. Table 12 gives the SIF and SIF+ABTT cells at train-selected layers, with intervals. Per-layer values for every setting in this appendix are in the code release.

Whitening. Full-rank PCA whitening (Su et al., 2021) is ill-conditioned here (768 to 1,024 dimensions estimated from 847 training vectors; condition numbers 3×10^5 to 10^{16}), and it ranks below ABTT at every panel layer. It also routes all but at most one test witness to an existing directory, so its assignment accuracy sits at the 62.4 percent class prior. Whitening to a reduced dimension, as Su et al. (2021) also do, drops the near-null directions and is the fairer comparison (Table 20). Fit on training embeddings with $r = 128$ or 256 components, it ranks on par with ABTT. At each model's train-selected layer, $r = 128$ reaches test AUROC 0.973–0.984, against 0.971–0.987 for ABTT. Over all 100 panel layers, collapsed layers included, both $r = 128$ and $r = 256$ stay within 0.021 of ABTT; $r = 64$ falls slightly below. On ranking, the repair therefore does not depend on ABTT in particular. On routing it does. At each model's train-selected routing layer, $r = 128$ reaches DirAcc@1 1.9 to 4.5 points below ABTT, and $r = 256$ up to 15 points below. We therefore keep ABTT as the reference correction.

	Ranking AUROC						Routing DirAcc@1					Routing assignment				
			whitening, r					whitening, r					whitening, r			
Model	Base	ABTT	64	128	256	full	ABTT	64	128	256	full	ABTT	64	128	256	full
LaTa	0.938	0.971	0.969	0.973	0.983	0.872	86.1	81.2	83.1	77.0	41.1	88.5	84.7	86.2	80.7	62.5
PhilTa	0.939	0.983	0.974	0.977	0.979	0.854	88.3	83.0	86.4	83.7	41.0	91.1	86.5	88.8	86.8	62.5
mT5-base	0.838	0.975	0.971	0.979	0.974	0.845	88.5	78.9	85.8	83.8	41.0	90.3	82.6	88.6	87.2	62.5
LaBSE	0.956	0.987	0.979	0.984	0.976	0.866	88.5	86.1	84.0	73.5	40.3	90.8	88.8	86.7	78.0	62.5
Qwen3-0.6B	0.966	0.973	0.973	0.978	0.983	0.594	89.4	84.8	85.0	75.9	20.5	91.5	87.2	88.0	79.3	62.4
KaLM-mini	0.972	0.981	0.976	0.981	0.981	0.562	89.4	82.6	86.7	84.5	16.8	91.7	85.4	89.2	87.8	62.4

Table 20: PCA whitening to r components against the baseline and ABTT, fit on training embeddings, at each setting's train-selected layer. Whitening keeps the top r principal components, rescaled to unit variance; full keeps all, at most 847 (the number of training passages). Routing in percent. Whitening ranks on par with ABTT, but routes below it.

J Cluster Geometry

This appendix shows the effect of ABTT on the whole corpus for the three raw T5-family panel models. Figure 7 gives t-SNE (van der Maaten and Hinton, 2008) maps for LaTa, PhilTa and mT5-base, each at the layer chosen on the training split by DirAcc@1 under ABTT, the routing ABTT layer of Table 16. The maps draw all 1,705 witnesses, color the same six largest directories, and compare baseline mean pooling with ABTT alone at $D = 10$. ABTT draws the witnesses of each colored directory into tighter groups than baseline mean pooling does. The maps are two-dimensional, so we read them as illustrations and compute no statistic on them.

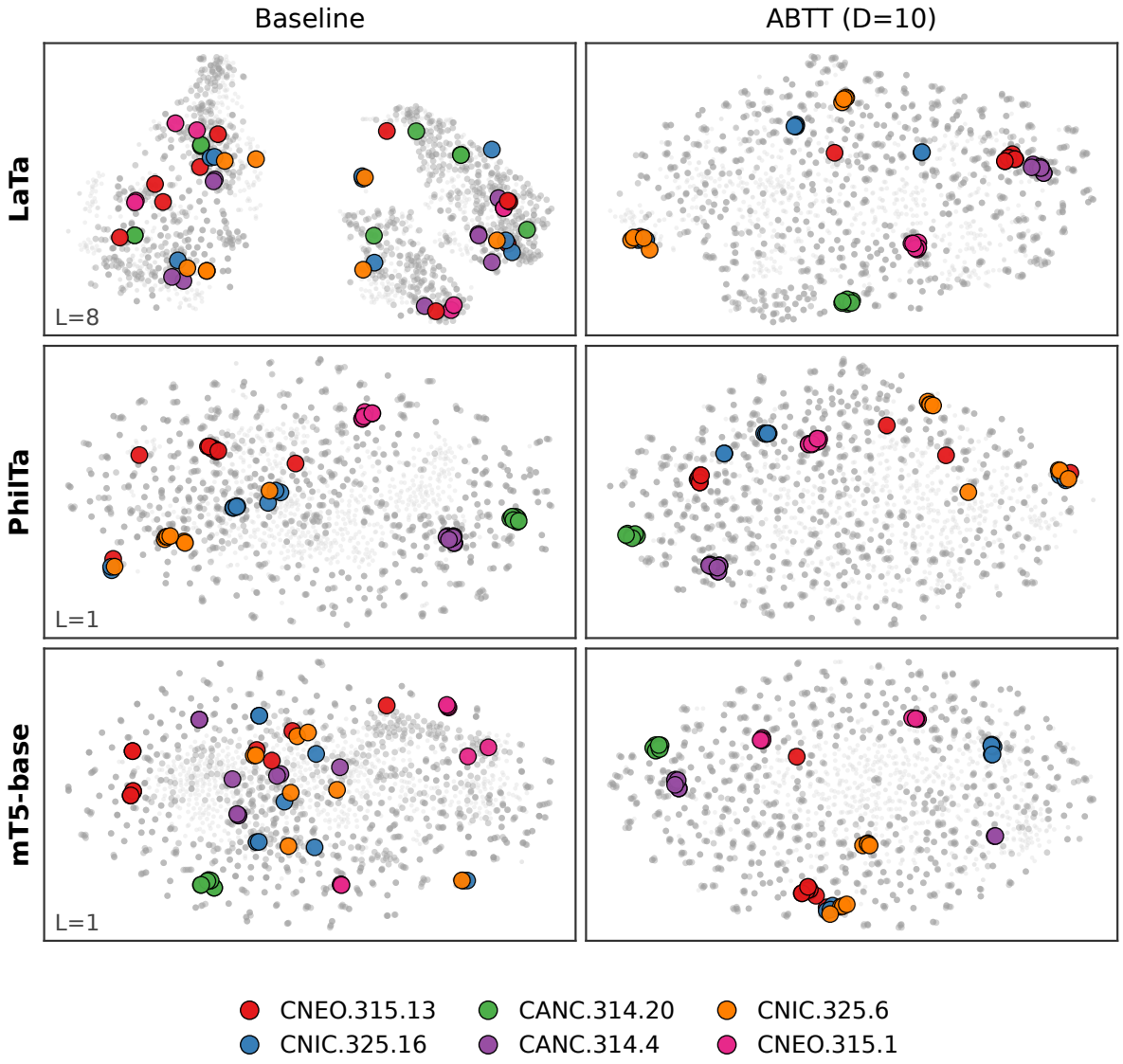


Figure 7: t-SNE maps of all 1,705 witnesses for LaTa, PhilTa and mT5-base at the train-selected routing ABTT layer (L; Table 16), under baseline mean pooling and ABTT alone ($D = 10$, fit on training embeddings). Mid gray: other non-singleton directories; light gray: singletons. ABTT draws each colored directory into a tighter group.